Passive Fault Tolerance through Tension-to-Thrust Feed-Forward: Hybrid Input-to-State Stability for Decentralized Multi-UAV Slung-Load Transport under Abrupt Cable Severance

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***Abstract*—Cooperative aerial transport with cable-suspended payloads must tolerate abrupt cable severance on the payload-pendulum timescale, which leaves insufficient time for fault detection, diagnosis, or reconfiguration. For this class of structural faults, a simple decentralized feed-forward identity sets each drone's altitude thrust feed-forward equal to its measured cable tension and provides the dominant passive fault-response mechanism. This identity converts the self-announcing change in load distribution into an immediate thrust adjustment using only local measurements and requires no communication or fault logic.**

We formalize this mechanism through a hybrid practical input-to-state stability analysis of the decentralized closed loop. We establish a reduced-order taut-cable model with explicit error bounds and construct a hybrid Lyapunov argument that yields an explicit recovery envelope with exponential decay and per-fault-cycle contraction. These results guarantee bounded tracking error and recovery across sequences of unannounced cable severances under actuator and dwell-time conditions.

High-fidelity multibody simulations demonstrate that the controller absorbs single and multiple severance events within one payload-pendulum period. Feed-forward ablations support the interpretation that the identity is the active recovery mechanism within the tested baseline architecture, while additional control layers remain inactive on the demonstrated trajectory family. These results show that, on the stated admissibility domain, passive local control laws can certify recovery for self-announcing cable-severance transitions without detection or reconfiguration.

***Index Terms*—Cooperative aerial manipulation, cable-suspended payload transport, decentralized fault-tolerant control, passive fault tolerance, hybrid practical input-to-state stability, Lyapunov contraction analysis, L_1 adaptive augmentation, receding-horizon model predictive control, anti-swing control, structural-transition systems.**

I. Introduction

Cooperative aerial transport uses multiple uncrewed aerial vehicles (UAVs) to carry a shared payload through flexible cables. In practical deployments, cable severance can occur due to abrasion, hardware fatigue, or obstacle contact. A severance event changes the constraint structure of the system, redistributes load across the remaining cables, and requires a compensatory response within one payload-pendulum period. This timescale is on the order of seconds and is shorter than the latency of conventional fault-tolerant control pipelines based on detection, classification, and reconfiguration.

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For this class of structural faults, a simple decentralized feed-forward identity enables passive fault tolerance. Each drone i sets its altitude thrust feed-forward equal to its measured cable tension.

$$T_i^{\mathrm{ff}}(t) = T_i(t). \tag{1}$$

When a cable severs, the resulting load redistribution appears immediately in the local tension measurement. The controller converts this change into a proportional thrust adjustment at the next control update using only local information. The mechanism requires no fault detection, no communication between agents, and no reconfiguration of the control law.

We analyze the resulting closed loop as a hybrid system with discrete structural transitions and continuous-time dynamics. The feed-forward identity supplies the bounded jump mechanism required for hybrid practical input-to-state stability, which enables recovery across sequences of unannounced cable severances under actuator and dwell-time conditions. The analysis yields an explicit recovery envelope with exponential decay and per-fault-cycle contraction.

We validate the proposed controller in high-fidelity multibody simulation across a range of fault scenarios. The results show that the system absorbs single and multiple cable severances within one payload-pendulum period. Feed-forward ablations support the interpretation that the identity is the active recovery mechanism within the tested baseline architecture, while additional control layers remain inactive on the demonstrated trajectory family.

This subject has not received the attention it deserves. Related work falls into four categories: centralized and distributed cooperative transport, geometric tracking and slung-load control, L_1 adaptive control, and fault-tolerant control.

Centralized and distributed cooperative transport is a force-allocation architecture [1]–[3] and distributed predictive variants [4]–[7] share a synchronous global-state requirement: the per-tick communication dependency is precisely what a cable-severance transition breaks at the actuator-tick rate, and none of these architectures accounts for cable severance in its stability argument. *Geometric tracking* [8], *flatness-based slung-load* [9], and $\mathrm{SE}(3)$ *payload control* [10] assume a fixed constraint set; multi-vehicle extensions [11], [12] reintroduce global-state exchange, collapsing the strict locality the present architecture requires. L_1 *adaptive control* provides matched-disturbance rejection with transient bounds [13], [14]; its extension to cooperative cable-fault settings under strict locality remains open [15].

The fault-tolerant control literature is divided into two main categories: active architectures, which reconfigure themselves after diagnosing a fault [16], [17], and passive architectures [16], [18], which maintain robustness to absorb faults without reconfiguration [19]. In the context of cooperative transport, approaches such as observer-based disturbance estimation with dynamic load reassignment address cable failures by detecting them and reconfiguring the system. Other methods, such as passive force control and leader-follower force estimation [20], avoid the need to exchange information between UAVs but rely on a fixed number of cables. Recent work has explored structured representations of system behavior and constraint prioritization as an alternative to reactive or detection-based approaches. [21]

Existing approaches either rely on detection and reconfiguration or assume a fixed constraint set. None provide a stability certificate for decentralized control under unannounced structural transitions with strict locality. In contrast, the present formulation identifies a specific feed-forward identity that enables passive fault tolerance and provides a rigorous hybrid stability analysis to certify its effectiveness under explicit conditions. The contribution is not just the mechanism itself, but the demonstration that it can be certified to work across structural transitions without detection or communication. The literature overlooks the following important aspects.

Why measured-force feedback and passive force control are insufficient? Measured-force feedback and passive impedance/admittance control are mature techniques for modulating stiffness or damping in fixed-topology systems. However, cable severance is not a disturbance on a fixed constraint structure; it is a *structural transition* that changes the cardinality and topology of the constraint set. Conventional passive force-control architectures do not address this class of faults because they assume a fixed number of cables and do not account for the discrete dynamics of a structural transition. Measured tension alone cannot induce a stabilizing response across a structural transition without *explicit certification* that the combined hybrid discrete-continuous dynamics remains stable. The present formulation addresses this gap: the identity $T_i^{\mathrm{ff}} = T_i$ is not merely a measured-force gain; it is a *mechanism for bounded-jump stabilization* in a hybrid system. The key contribution is the hybrid practical-ISS certificate that relates the identity to post-fault recovery through the payload-pendulum timescale constraint and the dwell hypothesis $\tau_d \geq \tau_{\mathrm{pend}}$. Without this certificate, the identity is heuristic; with it, the response to cable severance becomes predictable and auditable.

Why strict locality and no-communication decentralization matter? The fault-tolerant-control literature is accustomed to active reconfiguration: when a component fails, a centralized or coordinated mechanism detects it, classifies the fault, and reconfigures the control law or resource allocation. This pipeline requires communication and consensus, which takes time—time unavailable on the payload-pendulum period. The present approach abandons the detection-classification-reconfiguration pipeline in favor of a *self-announcing mechanism*: cable severance immediately changes the load distribution, and each drone's local tension sensor observes this change within one or two control ticks. By coupling this local observation directly to thrust, without peer-state exchange or a centralized diagnosis, the system responds passively and with zero latency outside the plant's native bandwidth. This eliminates the temporal gap between fault occurrence and response onset. The trade-off is that strict locality and no-communication prevent global optimization (such as ideal load sharing among survivors) and require a conditional certificate rather than a uniform one. The analysis certifies the decentralized no-communication response under explicit admissibility conditions (H1–H3), which the simulation campaign audits for the demonstrated mission family.

Three contributions organize the results for decentralized fault-tolerant control in cooperative aerial transport:

First, we introduce a strictly local tension-to-thrust feed-forward identity, $T_i^{\mathrm{ff}} = T_i$, for passive cable-severance tolerance in cooperative aerial transport. The identity converts the self-announcing redistribution of cable tension after severance into an immediate thrust correction, without fault detection, inter-agent communication, or controller reconfiguration. A reduced-order taut-cable model with explicit error bounds (Theorem 1) supplies the plant on which the identity is analyzed. (C1, Section II-A)

Second, we prove a conditional hybrid practical input-to-state stability certificate for the resulting decentralized closed loop. The proof combines a slack-excursion-bounded taut-cable reduction, inter-fault ISS decay, bounded fault jumps, and dwell-cycle contraction under explicit actuator and dwell-time conditions, yielding an explicit recovery envelope with exponential decay and per-fault-cycle contraction $\rho \in (0, 1)$. (C2, Section IV)

Third, we validate the mechanism in high-fidelity multi-body simulation under single- and dual-cable severance. Feed-forward ablations show that disabling $T_i^{\mathrm{ff}} = T_i$ inflates cruise RMSE by 34–39% and peak post-fault sag by 3.6–4.0×, supporting mechanism attribution to the identity within the tested baseline architecture. Extension layers (L_1, MPC, reshape) preserve the certificate when active and remain inactive on the demonstrated trajectory family. (C3, Section VI)

II. Problem Statement

We study cooperative aerial transport under unannounced cable-severance faults and formalize the plant, fault model, admissible controller class, reference trajectory, taut-cable reduction, and formal problem. The system consists of:

a) Vehicles, payload, and state.: A team of N quadrotor UAVs, indexed $i \in \mathcal{N} \triangleq \{1, \ldots, N\}$, operates in an inertial world frame $\mathcal{W}$. Drone i has mass m_i, diagonal inertia $\boldsymbol{J}_i$, position $\boldsymbol{p}_i$, velocity $\boldsymbol{v}_i$, attitude $\boldsymbol{R}_i \in \mathrm{SO}(3)$, and angular velocity $\boldsymbol{\omega}_i$, with per-drone state $x_i \triangleq (\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{R}_i, \boldsymbol{\omega}_i)$. A point-mass payload of mass m_L has state $x_L \triangleq (\boldsymbol{p}_L, \boldsymbol{v}_L)$; payload rotation is neglected.

b) Cables and complementarity condition.: Each drone i is connected to the payload by a flexible cable. Cable i exerts a world-frame force $\boldsymbol{T}_i^{\mathcal{W}} \in \mathbb{R}^3$ on the payload-side attachment point and an equal and opposite reaction on the drone-side

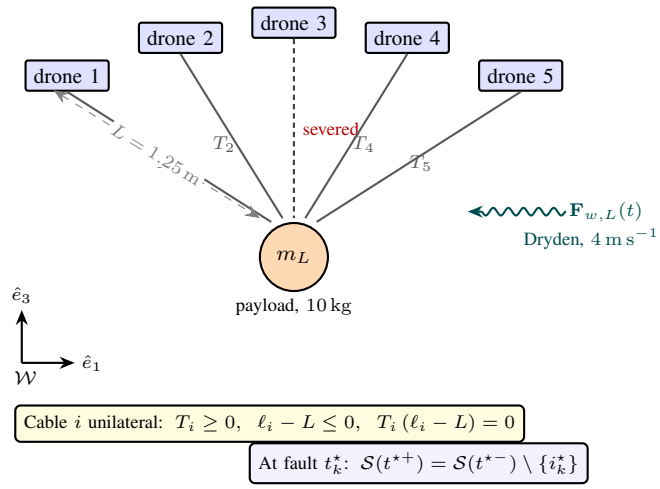


Fig. 1. Cooperative transport configuration. Five quadrotors carry a 10 kg point-mass payload through unilateral cables of rest length $L = 1.25\,\mathrm{m}$. Each drone i measures only its own state, its own scalar tension T_i, and the locally observed payload velocity $\boldsymbol{v}_L$; no peer drone or peer-cable signal is exchanged. The plant is driven by a bounded Dryden wind realisation ($4\,\mathrm{m\,s^{-1}}$ mean) and an unannounced cable-severance fault schedule $\sigma \in \mathcal{F}$ that decrements the survivor set $\mathcal{S}(t)$ at each event $t_k^\star$ (here drone $i^\star$, shown dashed).

attachment point. The cable transmits tensile force only when taut; its scalar tension is

$$T_i \triangleq \left\|\boldsymbol{T}_i^{\mathcal{W}}\right\| \geq 0, \tag{2}$$

with the unilateral constraint $T_i = 0$ whenever the cable is slack. The complementarity condition

$$T_i \geq 0, \quad \ell_i - L \leq 0, \quad T_i\left(\ell_i - L\right) = 0, \tag{3}$$

where $\ell_i = \|\boldsymbol{p}_{a,i} - \boldsymbol{p}_{b,i}\|$ is the cable's instantaneous length, $\boldsymbol{p}_{a,i}$ and $\boldsymbol{p}_{b,i}$ are the drone- and payload-side attachment points, and L is the rest length, captures the unilateral tensile constraint for an idealized inextensible cable. In the simulator each cable is a Kelvin–Voigt spring ($L = 1.25\,\mathrm{m}$, $k_s = 25\,000\,\mathrm{N\,m^{-1}}$); the elastic correction to the inextensible model is $O(\delta)$ and is absorbed in Theorem 1.

c) Equations of motion.: The drone translational dynamics

$$m_i\,\dot{\boldsymbol{v}}_i = f_i\,\boldsymbol{R}_i\,\hat{\boldsymbol{e}}_3 - m_i g\,\hat{\boldsymbol{e}}_3 - \boldsymbol{T}_i^{\mathcal{W}} + \boldsymbol{F}_{\mathrm{w},i}(t) \tag{4}$$

combine thrust $f_i \in [f_{\min}, f_{\max}]$, gravity, cable reaction, and wind. Standard rigid-body rotational dynamics $\boldsymbol{J}_i\dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times \boldsymbol{J}_i\boldsymbol{\omega}_i = \boldsymbol{\tau}_i$, $\dot{\boldsymbol{R}}_i = \boldsymbol{R}_i[\boldsymbol{\omega}_i]_\times$ govern attitude. The payload obeys

$$m_L\,\dot{\boldsymbol{v}}_L = -m_L g\,\hat{\boldsymbol{e}}_3 + \sum_{i\in\mathcal{S}(t)} \boldsymbol{T}_i^{\mathcal{W}} + \boldsymbol{F}_{\mathrm{w},L}(t), \tag{5}$$

with $\mathcal{S}(t) \subseteq \mathcal{N}$ the surviving-cable set. Per-drone inputs are $u_i = (f_i, \boldsymbol{\tau}_i) \in \mathcal{U}_i \triangleq [f_{\min}, f_{\max}] \times [-\tau_{\max}, \tau_{\max}]^3$, the plant state $x = (x_{\mathcal{N}}, x_L)$ lives in $\mathcal{X}$, and the admissible disturbance set is

$$\mathcal{D} \triangleq \left\{d(\cdot) : \|\boldsymbol{F}_{\mathrm{w},i}(t)\|,\ \|\boldsymbol{F}_{\mathrm{w},L}(t)\| \leq W_{\max},\ \forall i, t\right\}, \tag{6}$$

with $W_{\max} = 1\,\mathrm{N}$; the Dryden realization used in V2–V6 stays inside this bound (Table III).

At a time $t_k^\star$, cable $i_k^\star \in \mathcal{N}$ severs instantaneously. This structural fault event permanently removes cable $i_k^\star$ from the constraint set:

$$T_{i_k^\star}(t) = 0 \qquad \forall\, t \geq t_k^\star. \tag{7}$$

The active cable set transitions discretely:

$$\mathcal{S}(t_k^{\star,+}) = \mathcal{S}(t_k^{\star,-}) \setminus \{i_k^\star\}, \tag{8}$$

reducing the cardinality from $|\mathcal{S}|$ to $|\mathcal{S}| - 1$. For an elastic cable, the kinematic state $(\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{R}_i, \boldsymbol{\omega}_i, \boldsymbol{p}_L, \boldsymbol{v}_L)$ is continuous across severance; only the ODE right-hand side changes discontinuously at $t_k^\star$.

We consider a sequence of $F \leq N - 2$ fault events at times $\{t_k^\star\}_{k=1}^F$ with $t_1^\star < \cdots < t_F^\star$. The bound $F \leq N - 2$ keeps at least two cables intact; actuator margins limit the demonstrated scenario ($N = 5$, $m_L = 10\,\mathrm{kg}$) to $F \leq 2$. The inter-fault interval $\tau_{d,k} \triangleq t_{k+1}^\star - t_k^\star$ is the *dwell time*. We assume each fault is separated from the previous by at least one payload-pendulum period:

$$\tau_{d,k} \geq \tau_{\mathrm{pend}} \triangleq 2\pi\sqrt{L/g}, \qquad k = 1, \ldots, F-1, \tag{9}$$

reflecting the physical requirement that load redistribution from fault k settle before fault $k+1$. At $L = 1.25\,\mathrm{m}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$, $\tau_{\mathrm{pend}} \approx 2.24\,\mathrm{s}$. The admissible fault-schedule set is

$$\mathcal{F} \triangleq \Big\{\{(t_k^\star, i_k^\star)\}_{k=1}^F : F \leq N-2, \quad t_{k+1}^\star - t_k^\star \geq \tau_{\mathrm{pend}},\ i_k^\star \in \mathcal{S}(t_k^{\star,-})\Big\}, \tag{10}$$

$F = 0$ corresponds to the no-fault case.

d) Admissible controller class.: No fault flag, annunciation, or detection signal is transmitted; the fault manifests through a rise in $T_i(t)$ and a transient payload-velocity deviation, both in $\mathcal{I}_i(t)$. Field-deployable avionics impose a strict *admissible information pattern*. Drone i's controller π_i is restricted to the information set

$$\mathcal{I}_i(t) \triangleq \left\{x_i(t),\ T_i(t),\ \boldsymbol{v}_L(t),\ \boldsymbol{p}_L^d(\cdot),\ \boldsymbol{v}_L^d(\cdot)\right\}, \tag{11}$$

comprising the drone's inertial state x_i, its rope-tension measurement T_i, the payload velocity $\boldsymbol{v}_L$ (measured via a co-located downward-facing optical-flow sensor, no drone-to-drone communication), and a pre-distributed reference $(\boldsymbol{p}_L^d, \boldsymbol{v}_L^d)$ loaded before flight. Payload position $\boldsymbol{p}_L$ is deliberately excluded; only velocity enters the closed loop, via the anti-swing slot shift of Section III. Each π_i is a causal feedback map,

$$u_i(t) = \pi_i\Big(\mathcal{I}_i|_{[0,t]}\Big) \in \mathcal{U}_i, \tag{12}$$

possibly with a finite-dimensional internal state, but never with access to information outside (11). The admissible controller class is denoted Π_{loc}. $\mathcal{I}_i(t)$ excludes peer state, peer tension, fault flags, and run-time inter-drone packets; each π_i uses only co-located sensors and a pre-flight plan.

A. Reference Trajectory and Taut-Cable Reduction

The desired payload motion is a three-dimensional lemniscate trajectory with a sinusoidal altitude variation,

$$\boldsymbol{p}_L^d(t) \;=\; \begin{bmatrix} \dfrac{a\cos\phi(t)}{1+\sin^2\phi(t)} \\ \dfrac{a\sin\phi(t)\cos\phi(t)}{1+\sin^2\phi(t)} \\ z_0 + h_z\sin(\phi(t)) \end{bmatrix}, \phi(t) = \frac{2\pi(t-t_0)}{T_{\mathrm{ref}}}, \quad (13)$$

where $a = 3.0\,\mathrm{m}$ is the horizontal scale, $T_{\mathrm{ref}} = 12\,\mathrm{s}$ the period, $z_0 = 3.0\,\mathrm{m}$ the cruise altitude, and $h_z = 0.35\,\mathrm{m}$ the altitude excursion amplitude. These parameters yield a peak centripetal load of $\approx 0.25g$ at the figure-8 tips.

Each rope of rest length L is discretized into $N_b = 8$ identical Kelvin–Voigt beads of mass m_{bead} connected by $N_{\mathrm{seg}} = N_b + 1 = 9$ segments of stiffness k_s and damping c_s; N_{seg} is reserved for rope segments and is distinct from the surviving-cable cardinality $N_s(t) = |\mathcal{S}(t)| = N - F(t)$ used in Section IV. The distributed dynamics evolve on a fast timescale,

$$\tau_{\mathrm{rope}} \;=\; 2\pi\sqrt{m_{\mathrm{bead}}/k_s} \;\approx\; 6.3\,\mathrm{ms}, \quad (14)$$

separated from the payload-pendulum timescale $\tau_{\mathrm{pend}} \approx 2.24\,\mathrm{s}$ by $\delta = \tau_{\mathrm{rope}}/\tau_{\mathrm{pend}} \approx 0.003$. On the slow manifold, the net tensile force at the drone-side attachment equals the scalar $\xi_i = k_{\mathrm{eff}}(\ell_i - L)^+$, with effective stiffness

$$k_{\mathrm{eff}} \;=\; \frac{k_s}{N_{\mathrm{seg}}} \;=\; \frac{25\,000\,\mathrm{N\,m^{-1}}}{9} \;\approx\; 2778\,\mathrm{N\,m^{-1}}. \quad (15)$$

The six pre-registered missions (V1–V6) keep the system inside $\Omega_\tau^{\mathrm{dwell}}$ with margin (worst slack run $35.8\,\mathrm{ms}$, worst duty cycle 1.66%, Table V). The reduction replaces the $3N_b$-dimensional bead-chain state with the scalar ξ_i; the per-instant taut error is $O(\delta)$ and slack excursions add $O(\eta_{\max})$.

Theorem 1 (Taut-cable reduction)**.** *On the slack-excursion-bounded domain*

$$\Omega_\tau^{\mathrm{dwell}} \;\triangleq\; \big\{x \in \mathcal{X} : \textit{every slack run} \le \tau_{\mathrm{slack,max}}, \textit{ duty cycle} \le \eta_{\max}\big\}, \quad (16)$$

with $\tau_{\mathrm{slack,max}} = 40\,\mathrm{ms}$ and $\eta_{\max} = 2.5\%$, the distributed Kelvin–Voigt rope dynamics are approximated by the lumped scalar model with closed-loop shape-mode deviation

$$\|T_i^{\mathrm{true}} - T_i^{\mathrm{qs}}\| \;\le\; C_1\,\delta + C_2\,\eta_{\max}, \quad (17)$$

for constants $C_1, C_2 > 0$ depending on the bead-chain parameters $(m_{\mathrm{bead}}, c_s, k_s, N_b)$ and the wind bound $W_{\max}$, where $T_i^{\mathrm{qs}} \triangleq k_{\mathrm{eff}}(\ell_i(t) - L)^+$ is the rope-specific *quasi-static tension evaluated at the* same *current chord length $\ell_i(t)$ as T_i^{true}. This bound is on the shape-mode deviation of rope i from its own quasi-static approximation; it is distinct from the inter-rope tension asymmetry $\varepsilon_i = T_i - T_{\mathrm{eff}}$ reported in the simulation audit (Section VI-A), which is driven by formation geometry and can be $O(10)\,N$. The reduced-order model is therefore admissible as the plant for the stability analysis of Section IV.*

Sketch. The proof has two parts corresponding to the two terms in (17).

Part 1 ($C_1\delta$, taut intervals). Let $\tilde{\mathbf{q}} \in \mathbb{R}^{N_b}$ collect bead deviations from the quasi-static slow-manifold equilibrium $\bar{\mathbf{q}}(t)$ along the rope chord. Linearization yields the shape-mode dynamics $M_b\ddot{\tilde{\mathbf{q}}} + C_b\dot{\tilde{\mathbf{q}}} + K_b\tilde{\mathbf{q}} = -M_b\ddot{\bar{\mathbf{q}}}$ with C_b, K_b the chain damping and stiffness matrices. Differentiating the shape-mode energy $W = \frac{1}{2}\dot{\tilde{\mathbf{q}}}^\top M_b\dot{\tilde{\mathbf{q}}} + \frac{1}{2}\tilde{\mathbf{q}}^\top K_b\tilde{\mathbf{q}}$ and applying Young's inequality gives $\dot{W} \le -\alpha_b W + D_b$ with $\alpha_b = \lambda_{\min}(C_b)/m_{\mathrm{bead}}$ and $D_b = O(\delta^2)$ via the Tikhonov scaling $\|\ddot{\bar{\mathbf{q}}}\| = O(\delta)$. Gronwall yields $W^\infty = D_b/\alpha_b = O(\delta^2)$; the drone-side tension deviation $T_i^{\mathrm{true}} - T_i^{\mathrm{qs}} = k_s\mathbf{e}_1^\top\tilde{\mathbf{q}}$ then satisfies $\|T_i^{\mathrm{true}} - T_i^{\mathrm{qs}}\| \le C_1\delta \approx 0.072\,\mathrm{N}$ after the $\sim 6\,\mathrm{ms}$ transient. The slowest mode has $\zeta = 1.2$, $s_1 \approx -538\,\mathrm{s}^{-1}$.

Part 2 ($C_2\eta_{\max}$, slack intervals). On any slack run of duration $\le \tau_{\mathrm{slack,max}}$, the chord shortens by $\Delta\ell_{\mathrm{sag}}$ and the drone-side end segment must re-elongate on re-tension, producing a re-tension overshoot bounded by $k_{\mathrm{eff}}\Delta\ell_{\mathrm{sag}}$. Aggregating over the slack-duty budget gives $C_2 \approx 0.22\,\mathrm{N}$. At the worst observed $\eta_{\max} = 1.66\%$, the combined bound is $C_1\delta + C_2\eta_{\max} \approx 0.076\,\mathrm{N}$ per rope ($< 0.4\%$ of nominal per-drone tension). Both constants depend only on $(m_{\mathrm{bead}}, c_s, k_s, N_b)$ and $W_{\max}$. Detailed Gronwall arithmetic and shape-mode constants are in the supplementary material. □

Table V confirms all six missions reside inside $\Omega_\tau^{\mathrm{dwell}}$ with margin on every gate: worst slack run $35.8\,\mathrm{ms}$ ($4.2\,\mathrm{ms}$ headroom), duty cycle 1.66% (0.85 pp headroom), QP-transition fraction 0.10%. The theorem is asserted on the demonstrated regime, not an untested all-taut idealization.

The per-drone controller and the lemma stack in Section IV are structurally independent of N. The team size enters only through the post-fault redistributed load

$$T^{\mathrm{post}}(N, F) = \frac{m_L g}{N - F}, \quad (18)$$

yielding the actuator-margin condition

$$\frac{m_L g}{N - F} \le \kappa_{\mathrm{act}}\, f_{\max}, \quad (19)$$

with $\kappa_{\mathrm{act}} \in (0, 1)$ reserving headroom for tilt projection, anti-swing residual, and dynamic centripetal load. At canonical $\kappa_{\mathrm{act}} = 0.82$, $f_{\max} = 150\,\mathrm{N}$, and the demonstrated $(N, m_L, F) = (5, 10\,\mathrm{kg}, 2)$, (19) reads $32.7\,\mathrm{N} \le 123\,\mathrm{N}$, i.e. 27% of envelope. The certificate of Section IV carries unchanged at any (N, m_L, F) satisfying (19).

With the plant, fault model, reference trajectory, and reduction in place, we state the formal problem. Unilateral cable coupling (3), discrete structural fault transitions (7)–(8), admissible disturbances (6), and strict locality (11) together yield a hybrid decentralized control problem outside standard cooperative-transport formulations. The design variable is not a centralized control sequence but a family of local causal feedback laws. The problem, posed on the reduced-order model certified by Theorem 1, is as follows.

Definition 1 (Fault-tolerant decentralized aerial transport)**.** *Given the plant* (4)–(5)*, the complementarity constraint* (3)*, the mission reference* (13)*, all $d(\cdot) \in \mathcal{D}$* (6)*, and all $\sigma \in \mathcal{F}$* (10)*, find a causal controller family $\{\pi_i\}_{i=1}^N \in \Pi_{\mathrm{loc}}^N$ such that the closed loop satisfies the following.*

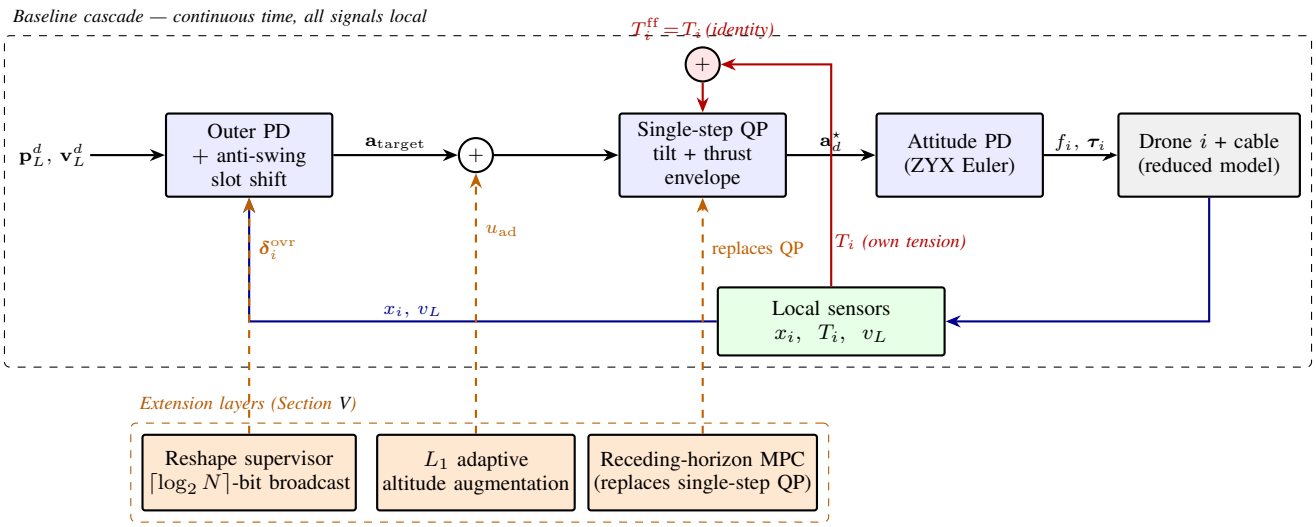


Fig. 2. Per-drone control cascade. The baseline (blue, framed top) is a three-stage continuous-time pipeline: outer-loop PD with anti-swing slot shift, a single-step convex QP that projects $\mathbf{a}_{\mathrm{target}}$ onto the tilt-and-thrust envelope, and an Euler-angle attitude PD. The non-standard element is the red feed-forward tap: drone i routes its own measured tension T_i directly into the altitude thrust feed-forward, $T_i^{\mathrm{ff}} = T_i$. This identity supplies the bounded-jump mechanism used in the hybrid practical-ISS certificate of Theorem 2; the surrounding cascade supplies inter-fault contraction and is otherwise textbook. Three extension layers (orange, framed bottom) hook in at three precisely identified injection points: the L_1 adaptive correction u_{ad} enters the altitude channel between the outer PD and the QP; the receding-horizon MPC replaces the single-step QP entirely; the reshape supervisor overrides the formation slot $\boldsymbol{\delta}_i$ on receipt of a single $\lceil \log_2 N \rceil$-bit broadcast.

P1 ***Nominal tracking.*** *The payload tracking error* $e_L(t) \triangleq \boldsymbol{p}_L(t) - \boldsymbol{p}_L^d(t)$ *is ultimately practically bounded,* $\limsup_{t\to\infty} \|e_L(t)\| \le \bar{e}_\infty + \bar{e}_0(W_{\max})$, *with* $\bar{e}_0 \in \mathcal{K}$.

P2 ***Passive fault absorption.*** *After any fault* $t_k^\star$ *obeying* (9), e_L *recovers to a bounded envelope,*

$$\|e_L(t)\| \le \beta\big(\|e_L(t_k^{\star,-})\|,\, t - t_k^\star\big) + \Gamma_f\, e^{-\lambda(t-t_k^\star)} + \bar{e}_\infty + \bar{e}_0(W_{\max}), \quad t \ge t_k^\star, \tag{20}$$

for some $\beta \in \mathcal{KL}$ *and constants* $\Gamma_f, \lambda > 0$, *with per-fault-cycle contraction* $V(x(t_{k+1}^{\star,-})) \le \rho V(x(t_k^{\star,-})) + c$ *for* $\rho \in (0,1)$ *and explicit* $c > 0$.

P3 ***Actuator feasibility.*** $f_i(t) \in [f_{\min}, f_{\max}]$ *and* $\boldsymbol{\tau}_i(t) \in [-\tau_{\max}, \tau_{\max}]^3$ *for all* $t \ge 0$ *and all admissible fault sequences, with non-zero margin.*

The discrete drop in $|\mathcal{S}|$ at each severance, the non-smooth taut-to-slack transitions inside $\Omega_\tau^{\mathrm{dwell}}$, and the strict locality constraint together require a hybrid Lyapunov argument on the tension mismatch

$$\boldsymbol{\varepsilon}_T(t) \triangleq \boldsymbol{T}(t) - \boldsymbol{T}^{\mathrm{qs}}(t), \tag{21}$$

developed in Section IV.

III. Proposed Method

Each drone runs a three-layer cascade: an outer-loop PD with anti-swing slot shift, a single-step convex QP on the tilt-and-thrust envelope, and an inner-loop attitude PD. The only nonstandard baseline element is the local feed-forward identity $T_i^{\mathrm{ff}} = T_i$, which maps tension redistribution after severance into immediate thrust compensation.

Each drone is pre-assigned a hovering slot displaced vertically above the payload reference point and laterally by a fixed formation offset,

$$\boldsymbol{p}_{\mathrm{slot},i}(t) = \boldsymbol{p}_L^d(t) + \boldsymbol{\delta}_i + d_{\mathrm{rope}}\,\hat{\boldsymbol{e}}_3, \tag{22}$$

where $\boldsymbol{\delta}_i \in \mathbb{R}^3$ is the formation offset (overridable at run time), $d_{\mathrm{rope}} = 1.25\,\mathrm{m}$ is the vertical slot elevation matching L, and $\hat{\boldsymbol{e}}_3$ is the upward unit vector. The slot velocity is $\boldsymbol{v}_{\mathrm{slot},i}(t) = \boldsymbol{v}_L^d(t)$. To damp the payload pendulum passively, we shift the slot relative to the payload's horizontal velocity. With $\boldsymbol{v}_{L,\perp} \triangleq [v_{L,x}, v_{L,y}, 0]^\top$, the horizontal swing correction is

$$\boldsymbol{s}_i(t) = \begin{cases} -k_{\mathrm{swing}}\,\boldsymbol{v}_{L,\perp} & \text{if } k_{\mathrm{swing}}\|\boldsymbol{v}_{L,\perp}\| \le s_{\max}, \\ -s_{\max}\,\hat{\boldsymbol{v}}_{L,\perp} & \text{otherwise}, \end{cases} \tag{23}$$

where $\hat{\boldsymbol{v}}_{L,\perp}$ is the unit vector of $\boldsymbol{v}_{L,\perp}$. The shifted slot position is $\boldsymbol{p}_{\mathrm{slot},i}^{\mathrm{dyn}}(t) = \boldsymbol{p}_{\mathrm{slot},i}(t) + \boldsymbol{s}_i(t)$. therefore, the position and velocity errors are

$$\begin{aligned} \boldsymbol{e}_p(t) &= \boldsymbol{p}_{\mathrm{slot},i}^{\mathrm{dyn}}(t) - \boldsymbol{p}_i(t), \\ \boldsymbol{e}_v(t) &= \boldsymbol{v}_{\mathrm{slot},i}(t) - \boldsymbol{v}_i(t). \end{aligned} \tag{24}$$

The outer loop computes a target body acceleration,

$$\boldsymbol{a}_{\mathrm{target}}(t) = \underbrace{\begin{bmatrix} K_p^{xy}\, e_{p,x} + K_d^{xy}\, e_{v,x} \\ K_p^{xy}\, e_{p,y} + K_d^{xy}\, e_{v,y} \\ K_p^{z}\, e_{p,z} + K_d^{z}\, e_{v,z} \end{bmatrix}}_{\boldsymbol{a}_{\mathrm{track}}} + w_{\mathrm{swing}} \begin{bmatrix} -k_{\mathrm{swing}}\, v_{L,x} \\ -k_{\mathrm{swing}}\, v_{L,y} \\ 0 \end{bmatrix}. \tag{25}$$

The altitude gain $K_p^z = 100$ exceeds the horizontal gain $K_p^{xy} = 30$ because vertical rope forces dominate the disturbance budget; the formal pole and Lyapunov-matrix analysis is

TABLE I
PROPOSED CONTROLLER PARAMETERS (CANONICAL VALUES USED IN ALL SIMULATION CAMPAIGNS).

Parameter	Value	Role
$K_p^{xy},\ K_d^{xy}$	30, 15	horizontal outer loop
$K_p^{z},\ K_d^{z}$	100, 24	altitude outer loop
$K_p^{\mathrm{att}},\ K_d^{\mathrm{att}}$	25, 4	attitude inner loop
k_{swing}	0.8	anti-swing gain
w_{swing}	0.3	anti-swing weight in $\boldsymbol{a}_{\mathrm{target}}$
$s_{\max}$	0.3 m	slot-shift saturation
d_{rope}	1.25 m	slot z-offset (nominal rope length)
$w_t,\ w_e$	1.0, 0.02	QP tracking / effort weights
$\theta_{\max}$	0.6 rad	tilt ceiling ($\approx 34^\circ$)
$f_{\min},\ f_{\max}$	0, 150 N	thrust envelope
$\tau_{\max}$	10 N m	torque saturation
$\bar{T}$	19.62 N	nominal per-drone tension ($m_L g/N$)

in Lemma 2. When L_1 is active, its altitude correction $u_{\mathrm{ad}}(t)$ is added to the z-component of $\boldsymbol{a}_{\mathrm{target}}$ between the outer PD and the QP (Section V); horizontal channels are unmodified.

The target acceleration $\boldsymbol{a}_{\mathrm{target}}$ may lie outside the tilt-and-thrust actuator envelope. We project onto the feasible set with a single-step convex QP that also regularises large commands:

$$\boldsymbol{a}_d^\star = \underset{\boldsymbol{a}_d \in \mathbb{R}^3}{\operatorname{argmin}}\ w_t \left\|\boldsymbol{a}_d - \boldsymbol{a}_{\mathrm{target}}\right\|^2 + w_e \left\|\boldsymbol{a}_d\right\|^2, \tag{26a}$$

$$\text{s.t. } |a_{d,x}|,\ |a_{d,y}| \le a_{\mathrm{tilt}} \triangleq g \tan\theta_{\max}, \tag{26b}$$

$$a_{z,\min}(t) \le a_{d,z} \le a_{z,\max}(t), \tag{26c}$$

where the vertical bounds incorporate the rope tension feed-forward,

$$a_{z,\min}(t) = \frac{f_{\min} - T_i^{\mathrm{ff}}(t)}{m_i} - g, \quad a_{z,\max}(t) = \frac{f_{\max} - T_i^{\mathrm{ff}}(t)}{m_i} - g. \tag{27}$$

The vertical interval in (26c) is non-empty whenever $f_{\min} < f_{\max}$; hover feasibility holds in all scenarios, and Doctrine D1 is satisfied at the 0.10% worst QP-transition fraction (Table V). For the canonical campaigns, the feed-forward is the identity from $t = 0$:

$$\boxed{T_i^{\mathrm{ff}}(t) \ = \ T_i(t).} \tag{28}$$

A smoothstep pickup ramp $\alpha(\tau) = 3\tau^2 - 2\tau^3$ handles cold-start engagement; it is bypassed in every reported campaign.

Given the QP solution $a_{d,z}^\star$, the commanded thrust is

$$f_i \ = \ \mathrm{clamp}\big(m_i(g + a_{d,z}^\star) + T_i^{\mathrm{ff}},\ f_{\min},\ f_{\max}\big). \tag{29}$$

Under the small-tilt regime enforced by (26b), the horizontal acceleration commands map to desired pitch and roll via $\theta_{\mathrm{pitch}}^d = \mathrm{sat}(a_{d,x}^\star/g, \pm\theta_{\max})$, $\theta_{\mathrm{roll}}^d = \mathrm{sat}(-a_{d,y}^\star/g, \pm\theta_{\max})$, tracked by a per-axis Euler-angle PD with gains $(K_p^{\mathrm{att}}, K_d^{\mathrm{att}})$ and torques saturated to $[-\tau_{\max}, \tau_{\max}]^3$. The yaw channel uses the standard $\mathrm{SO}(3)$-projected error $-(R_{21} - R_{12})/2$. Table I collects the canonical controller parameters used throughout all simulation campaigns.

IV. STABILITY ANALYSIS

The baseline controller of Section III achieves fault tolerance through a single structural choice: setting the altitude thrust feed-forward equal to the drone's own measured rope tension, $T_i^{\mathrm{ff}} = T_i$. We establish that this identity renders the closed loop *hybrid practical input-to-state stable* (hybrid practical-ISS) under the cable-severance fault model of Section II-A. When a peer cable severs, T_i rises immediately and the feed-forward converts that step into a proportional thrust increment at the plant's native rate, without any detection, annunciation, or reconfiguration interval.

The lemmas of Section IV-B verify the hybrid-systems hypotheses of [22] on $\Omega_\tau^{\mathrm{dwell}}$. The decay rates λ, ρ are explicit from the Lyapunov matrices P_v, P_{xy} in Lemma 2; the jump bound $\chi(\Delta_f)$ is structurally explicit via (37), with the tension-weight κ_V deferred to follow-on calibration without affecting the $\rho < 1$ certificate.

A. Setup, Notation, and Analytical Hypotheses

The analysis operates on the reduced-order model of Section II-A. Let $e_{p,z,i}(t) = p_{\mathrm{slot},i,z}(t) - p_{i,z}(t)$ be drone i's altitude slot-tracking error and $\boldsymbol{e}_p(t) \in \mathbb{R}^{3N_s}$ the stacked position error over $\mathcal{S}(t)$ with $N_s = |\mathcal{S}(t)|$. Consecutive faults are separated by at least one payload-pendulum period,

$$\tau_{d,k} \ \ge \ \tau_{\mathrm{pend}} \ \triangleq \ 2\pi\sqrt{L/g}. \tag{30}$$

At $L = 1.25\,\mathrm{m}$, $\tau_{\mathrm{pend}} = 2.24\,\mathrm{s}$ and the pendulum natural frequency is $\omega_p \triangleq \sqrt{g/L} = 2.80\,\mathrm{rad\,s^{-1}}$.

The analysis rests on three hypotheses, defended empirically in Tables II and V: **H1** (slack-excursion budget, $\tau_{\mathrm{slack,max}} \le 40\,\mathrm{ms}$ and $\eta_{\max} \le 2.5\%$); **H2** (spaced-fault dwell, $\tau_{d,k} \ge \tau_{\mathrm{pend}}$ (30)); **H3** (Doctrine D1, single-active-regime QP whose active set $\mathcal{A}(t)$ is locally constant inside each inter-fault window). The H1 gate bounds are set by bead-chain physics: $40\,\mathrm{ms}$ is one quarter of the Tikhonov frozen window $\tau_{\mathrm{pend}}/(2\pi)$, and 2.5% is a Rayleigh-turbulence budget. Theorem 1 is independently validated against the bead-chain truth model in Section VI-A, providing an *a priori* check that the reduction error is absorbed into γ_η of Theorem 2 without appeal to the simulated missions; the partial post-hoc nature of H1/H3 (defended on the same missions) is bounded by the audit margins of Table V.

B. Lemmas Underpinning the Hybrid Practical-ISS Theorem

Four lemmas plus the anti-swing invariance remark (Remark 1) underpin Theorem 2: tension cancellation on the altitude channel (Lemma 1); ISS decay for altitude and horizontal channels via Lyapunov matrices P_v, P_{xy} (Lemma 2); a bounded Lyapunov jump $\chi(\Delta_f)$ at each fault (Lemma 3); and the dwell-cycle contraction $\rho \in (0, 1)$ composing jump and decay (Lemma 4). Pre- and post-fault tension coordinates are compared on the surviving set by restricting the pre-fault vector to the remaining cables.

Lemma 1 (Tension cancellation)**.** *Assume H1 and the cable-tilt domain condition $|\theta_i(t)| \le \theta_{c,\max}$ for known $\theta_{c,\max} \in (0, \pi/2)$. Then $T_i^{\mathrm{ff}} = T_i$ cancels the dominant projection term $T_i \cos\theta_i$ from the drone altitude acceleration, leaving a residual factor $(1 - \cos\theta_i)$ bounded by $1 - \cos\theta_{c,\max}$. At $\theta_{c,\max} = 0.75\,\mathrm{rad}$ this is 0.27, and the residual enters*

the altitude-error equation as a bounded matched disturbance handled by Lemma 2. The bound holds uniformly in $N_s \geq 2$.

Proof. Substituting $f_i = m_i(g + a^\star_{d,z}) + T_i^{\mathrm{ff}}$ into $m_i\ddot{p}_{i,z} = f_i - m_i g - T_i \cos\theta_i + F_{w,i,z}$ and absorbing the wind term yields

$$m_i\ddot{p}_{i,z} = m_i a^\star_{d,z} + T_i(1 - \cos\theta_i). \tag{31}$$

Since $|\theta_i| \leq \theta_{c,\max}$ uniformly (enforced by the QP tilt ceiling), the residual $T_i(1 - \cos\theta_i) \leq T_i(1 - \cos\theta_{c,\max})$ regardless of N_s, and is absorbed by γ_q of Lemma 2. □

Lemma 2 (ISS decay via Lyapunov matrices)**.** *Under H1–H3, both the altitude error $e_{p,z,i}$ and horizontal error $e_{p,xy,i}$ satisfy exponential ISS between fault events. Let $A_m = \begin{bmatrix} 0 & 1 \\ -K_p^z & -K_d^z \end{bmatrix}$ and $A_{m,xy} = \begin{bmatrix} 0 & 1 \\ -K_p^{xy} & -K_d^{xy} \end{bmatrix}$ be the closed-loop matrices for gains $(K_p^z, K_d^z) = (100, 24)$ and $(K_p^{xy}, K_d^{xy}) = (30, 15)$. The unique solutions $P_v \succ 0$ and $P_{xy} \succ 0$ of $A_m^\top P_v + P_v A_m = -I$ and $A_{m,xy}^\top P_{xy} + P_{xy} A_{m,xy} = -I$ are*

$$P_v = \begin{bmatrix} 2.224 & 0.005 \\ 0.005 & 0.021 \end{bmatrix}, \qquad P_{xy} = \begin{bmatrix} 1.283 & 0.017 \\ 0.017 & 0.034 \end{bmatrix}. \tag{32}$$

Setting $V_z = \frac{1}{2}[e_{p,z}, \dot{e}_{p,z}]^\top P_v [e_{p,z}, \dot{e}_{p,z}]$ and analogously V_{xy}, there exist $\gamma_w, \gamma_q, \gamma_w^{xy}, \gamma_q^{xy} \in \mathcal{K}$ such that, between fault events,

$$\dot{V}_z \;\leq\; -\lambda_z V_z + \gamma_w(\|w\|_\infty) + \gamma_q(\|q - q^\star\|_\infty), \tag{33}$$

$$\dot{V}_{xy} \;\leq\; -\lambda_{xy} V_{xy} + \gamma_w^{xy}(\|w\|_\infty) + \gamma_q^{xy}(\|q - q^\star\|_\infty), \tag{34}$$

with $\lambda_z = \alpha_z = |\mathrm{Re}(s_1)| = 5.37\,\mathrm{s}^{-1}$ (dominant altitude pole $s_1 = -12 + \sqrt{44}$) and $\lambda_{xy} = \alpha_{xy} = |\mathrm{Re}(s_1^{xy})| = 2.38\,\mathrm{s}^{-1}$ (dominant horizontal pole $s_1^{xy} = (-15 + \sqrt{105})/2$). Since $\alpha_{xy} < \alpha_z$, the horizontal channel is the bottleneck: $\alpha_{\min} = \alpha_{xy} = 2.38\,\mathrm{s}^{-1}$, which drives the dwell-cycle contraction of Lemma 4. Both rates are independent of N_s.

Proof. After tension cancellation (Lemma 1), each scalar error satisfies a Hurwitz closed-loop ODE; for altitude:

$$\ddot{e}_{p,z,i} + K_d^z \dot{e}_{p,z,i} + K_p^z e_{p,z,i} = \ddot{p}_{\mathrm{slot},z} + r_i(t). \tag{35}$$

The unique $P \succ 0$ solving $A^\top P + PA = -I$ is computed algebraically, yielding the matrices in (32); differentiating $V = \frac{1}{2}\zeta^\top P \zeta$ along trajectories and applying Young's inequality gives (33)–(34) with rates $|\mathrm{Re}(s_1)|$. Independence from N_s follows because neither A_m nor $A_{m,xy}$ depends on $|\mathcal{S}|$. □

Remark 1 (Pendulum damping invariance)**.** *The anti-swing damping coefficient is N_s-independent: the restoring force $N_s T_i^{\mathrm{ss}} k_{\mathrm{swing}} v_{L,\perp}/L$ with $T_i^{\mathrm{ss}} = m_L g/N_s$ satisfies $N_s T_i^{\mathrm{ss}} = m_L g$, leaving $B_{\mathrm{swing}} = m_L g\, k_{\mathrm{swing}}/L = 62.8\,\mathrm{N\,s\,m^{-1}}$ and damping ratio $\zeta = B_{\mathrm{swing}}/(2 m_L \omega_p) = 1.12 > 1$ (over-damped) for any $N_s \geq 2$.*

Lemma 3 (Bounded fault jump with explicit χ)**.** *At each fault event t_f, the Lyapunov function undergoes a bounded jump,*

$$V(\boldsymbol{\xi}(t_f^+)) \;\leq\; V(\boldsymbol{\xi}(t_f^-)) \;+\; \chi(\Delta_f), \qquad \chi \in \mathcal{K}, \tag{36}$$

where $\boldsymbol{\xi} = (\boldsymbol{e}_p, \boldsymbol{e}_v, \boldsymbol{\varepsilon}_T)$, the pre-fault tension coordinates are restricted to the post-fault survivor set through the map $\Pi_{\mathcal{S}^+ \leftarrow \mathcal{S}^-}$, and $\Delta_f = \|\boldsymbol{T}^{\mathrm{qs}}(t_f^+) - \Pi_{\mathcal{S}^+ \leftarrow \mathcal{S}^-} \boldsymbol{T}^{\mathrm{qs}}(t_f^-)\|$ is the quasi-static tension redistribution over the surviving cables induced by the severance. On the admissible domain of H1–H3, the jump function admits the bound

$$\chi(\Delta_f) \;\leq\; \bar{c}_f\big(\Delta_f + \Delta_f^2\big), \tag{37}$$

for some $\bar{c}_f > 0$ depending only on the admissible domain and the Lyapunov weighting of the tension channel. Under symmetric equal sharing at hover, the linear term vanishes and (37) *reduces to $\chi(\Delta_f) \leq \kappa_V \Delta_f^2/(2N_s)$.*

Sketch. The Kelvin–Voigt cable model keeps $(\boldsymbol{p}_i, \boldsymbol{v}_i, p_L, v_L)$ continuous across t_f since no impulsive force is applied; on the severed rope, tension drops to zero within $\tau_{\mathrm{rope}} \approx 6.3\,\mathrm{ms}$ (14). Decompose $V = V_{\mathrm{track}} + \kappa_V V_{\mathrm{tension}}$. By kinematic continuity and pre-loaded slot reference, $V_{\mathrm{track}}(t_f^+) \leq V_{\mathrm{track}}(t_f^-)$. The survivor tensions $T_i = k_s(\ell_i - L)^+$ are continuous functionals of the (continuous) kinematic state, so $\boldsymbol{\varepsilon}_T(t_f^+) = \Pi_{\mathcal{S}^+ \leftarrow \mathcal{S}^-} \boldsymbol{\varepsilon}_T(t_f^-) - \boldsymbol{b}_f$ with $\|\boldsymbol{b}_f\| = \Delta_f$. Quadratic expansion of V_{tension} about $\Pi \boldsymbol{\varepsilon}_T(t_f^-)$ on the compact admissible domain yields $V_{\mathrm{tension}}(t_f^+) - V_{\mathrm{tension}}(t_f^-) \leq c_{f,1}\Delta_f + c_{f,2}\Delta_f^2$. Absorbing $\kappa_V c_{f,j}$ into $\bar{c}_f$ gives (37). Under the canonical symmetric hover ($\boldsymbol{\varepsilon}_T(t_f^-) = 0$) the linear term vanishes and survivor-count algebra reduces (37) to $\chi(\Delta_f) \leq \kappa_V \Delta_f^2/(2N_s)$. □

Lemma 4 (Dwell-cycle contraction)**.** *Let $t_k^\star$ and $t_{k+1}^\star$ be consecutive fault events separated by dwell-time $\tau_{d,k} \geq \tau_{\mathrm{pend}}$. Under the bounded fault jump of Lemma 3 and the ISS decays of Lemma 2, the Lyapunov function satisfies a per-fault-cycle contraction*

$$V(\boldsymbol{\xi}(t_{k+1}^{\star,-})) \;\leq\; \rho\, V(\boldsymbol{\xi}(t_k^{\star,-})) \;+\; c, \tag{38}$$

$$\rho \;\triangleq\; \exp(-\alpha_{\min}\tau_{\mathrm{pend}}), \quad \alpha_{\min} = \alpha_{xy} = 2.38\,\mathrm{s}^{-1}, \tag{39}$$

where

$$c \;=\; \rho\,\chi(\Delta_f) \;+\; \frac{c_0}{\alpha_{\min}}, \tag{40}$$

$$c_0 = \sup_t \big[\gamma_w(\|w(t)\|_\infty) + \gamma_q(\|q - q^\star\|_\infty) + \gamma_\eta(\eta_{\max})\big].$$

At canonical parameters, $\rho \approx \exp(-2.38 \times 2.24) \approx 0.005$.

Proof. Applying Lemma 3 at $t_k^\star$:

$$V(t_k^{\star,+}) \;\leq\; V(t_k^{\star,-}) + \chi(\Delta_f). \tag{41}$$

Between $t_k^{\star,+}$ and $t_{k+1}^{\star,-}$, the flow satisfies $\dot{V} \leq -\alpha_{\min} V + c_0$ (combining Lemma 2 through the composite Lyapunov function $V = V_z + V_{xy} + \kappa_V V_{\mathrm{tension}}$, using $\alpha_{\min} = \min(\alpha_z, \alpha_{xy}) = \alpha_{xy}$). Integrating over an interval of length $\tau \geq \tau_{\mathrm{pend}}$:

$$V(t_{k+1}^{\star,-}) \;\leq\; e^{-\alpha_{\min}\tau}\big(V(t_k^{\star,-}) + \chi(\Delta_f)\big) \;+\; \frac{c_0}{\alpha_{\min}}(1 - e^{-\alpha_{\min}\tau}). \tag{42}$$

Setting $\rho = e^{-\alpha_{\min}\tau_{\mathrm{pend}}}$ and upper-bounding τ by τ_{pend} (the minimum dwell, giving the worst-case exponential decay) yields (38)–(40). Since $\alpha_{\min} = \alpha_{xy} > 0$ and $\tau_{\mathrm{pend}} > 0$, we have $\rho \in (0, 1)$. □

Remark 2 (Analytical ρ vs. empirical $\hat{\rho}$)**.** *Lemma 4 gives $\rho_{\mathrm{ana}} = \exp(-\alpha_{xy}\tau_{\mathrm{pend}}) \approx 0.005$, while measured V4/V5*

Lyapunov-proxy ratios are $\hat{\rho}_{\mathrm{V4}} = 0.20, \hat{\rho}_{\mathrm{V5}} = 0.045$. The gap is the additive disturbance floor $c > 0$:

$$\hat{\rho}_k \triangleq \frac{V(t_{k+1}^{\star,-})}{V(t_k^{\star,-})} \leq \rho_{\mathrm{ana}} + \frac{c}{V(t_k^{\star,-})}, \tag{43}$$

tight only in the zero-disturbance limit. Under Dryden $4\,\mathrm{m\,s^{-1}}$ wind, $c/V(t_k^{\star,-}) \approx 0.195$ (V4) and ≈ 0.040 (V5); the V4→V5 improvement reflects further pre-fault decay over the longer dwell.

C. Main Theorem, Robustness, and Multi-Fault Feasibility

We now compose the five lemmas into the hybrid practical-ISS theorem and state corollaries on robustness, the steady-state tracking bound, and multi-fault feasibility.

Theorem 2 (Hybrid practical-ISS with explicit recovery envelope under F sequential agent failures)**.** *Under hypotheses H1–H3 (Section IV-A), the cable-tilt domain condition of Lemma 1, any fault schedule $\sigma = \{(t_k^\star, i_k^\star)\}_{k=1}^F \in \mathcal{F}$ of F unannounced cable severances satisfying $F \leq N-2$ and the per-fault actuator-margin condition $m_L g/(N-k) \leq \kappa_{\mathrm{act}} f_{\max}$ for all $k \in \{1,\ldots,F\}$, and admissible disturbances $d \in \mathcal{D}$, the stacked tracking-and-tension-error state $\boldsymbol{\xi} = (\boldsymbol{e}_p, \boldsymbol{e}_v, \boldsymbol{\varepsilon}_T)$ satisfies*

$$\begin{aligned}\|\boldsymbol{\xi}(t)\| \;\leq\; & \beta\big(\|\boldsymbol{\xi}(0)\|, t\big) + \gamma_w\big(\|w\|_\infty\big) + \gamma_q\big(\|q - q^\star\|_\infty\big) \\ & + \gamma_\eta\big(\eta\big) + \sum_{k=1}^{F} \Gamma_f\, e^{-\lambda(t-t_k^\star)}\, \mathbf{1}_{t \geq t_k^\star}, \\ & t \in [t_{\mathrm{pickup}}, T],\end{aligned} \tag{44}$$

where $\beta \in \mathcal{KL}$, $\gamma_w, \gamma_q, \gamma_\eta \in \mathcal{K}$, $\lambda \triangleq \alpha_{\min}/2 = \alpha_{xy}/2 = 1.19\,\mathrm{s}^{-1}$ is the norm-level recovery rate induced by the slower of the altitude and horizontal closed-loop spectral rates (Lemma 2), $\eta = \eta(\boldsymbol{\xi}[\cdot])$ is the realized slack duty cycle bounded by $\eta_{\max}$ via H1, and $\Gamma_f > 0$ is the fault-overshoot coefficient set by the redistribution geometry. Equivalently, $V(\boldsymbol{\xi}(t_{k+1}^{\star,-})) \leq \rho V(\boldsymbol{\xi}(t_k^{\star,-})) + c$ for explicit $c > 0$ per Lemma 4. Each surviving cable's tension obeys the practical envelope

$$\begin{aligned}|T_i(t) - T_i^{\mathrm{qs}}(t)| \;\leq\; & T_{\mathrm{overshoot}}\, e^{-\lambda(t-t_{\mathrm{last}}^\star)} + T_{\mathrm{floor}}, \\ & i \in \mathcal{S}(t),\end{aligned} \tag{45}$$

where $T_{\mathrm{overshoot}}$ is bounded by the corresponding tension-error radius extracted from (37)*, and T_{floor} is the steady-state floor induced by γ_w, γ_q, and γ_η.*

Sketch. Inter-fault ISS. On each inter-fault interval $[t_k^{\star,+}, t_{k+1}^{\star,-}]$, Lemma 1 reduces the altitude dynamics to (35) with bounded residual on $\Omega_\tau^{\mathrm{dwell}}$. Using the composite $V = \lambda_{\min}(P_v)^{-1} V_z + \lambda_{\min}(P_{xy})^{-1} V_{xy} + \kappa_V V_{\mathrm{tension}}$, Lemma 2 give $\dot{V} \leq -\alpha_{\min} V + \gamma_w + \gamma_q + \gamma_\eta$ with $\alpha_{\min} = \alpha_{xy}$ and γ_η absorbing the reduction residual of Theorem 1. The comparison lemma yields $\beta \in \mathcal{KL}$ and $\gamma_w, \gamma_q, \gamma_\eta \in \mathcal{K}$. Quadratic equivalence $V \sim \|\xi\|^2$ converts the Lyapunov decay rate to the norm rate $\lambda = \alpha_{\min}/2$ (see Remark 3).

Per-fault jump. Lemma 3 gives $V(t_k^{\star,+}) \leq V(t_k^{\star,-}) + \chi(\Delta_{f,k})$, and Lemma 4 composes jump and decay into $V(t_{k+1}^{\star,-}) \leq \rho V(t_k^{\star,-}) + c$. The exponential damping of the jump produces the $\Gamma_f e^{-\lambda(t-t_k^\star)}$ term in (44).

Induction over $F \leq N-2$ faults. Geometric summation gives $V(t_k^{\star,-}) \leq \rho^{k-1} V(t_1^{\star,-}) + c(1-\rho^{k-1})/(1-\rho)$, bounded as long as the actuator-margin condition $m_L g/(N-k) \leq \kappa_{\mathrm{act}} f_{\max}$ holds for each k. Composing through the hybrid-system framework [22, Thm. 1] yields (44); the tension envelope (45) follows from $V_{\mathrm{tension}} \sim \|\boldsymbol{\varepsilon}_T\|^2$ and $\varepsilon_{T,i} = T_i - T_i^{\mathrm{qs}}$. □

Remark 3 (Jump-bound vs. contraction-rate mechanism)**.** *The ablation (Section VI-D) shows that disabling $T_i^{\mathrm{ff}} = T_i$ increases peak sag by 3.6–4.0× and produces larger post-fault Lyapunov jumps (Figure 4), supporting the jump-bound role of the identity versus the contraction-rate role of α_{xy} within the tested architecture.*

Corollary 1 (Pre-fault steady-state tracking bound)**.** *Prior to any fault event, the payload tracking error is dominated by the quasi-static pendulum offset and the outer-loop PD residual:*

$$\bar{e}_{p,xy} \leq \frac{L A_{xy,\max}}{g}\left(1 - \frac{1}{2\zeta^2}\right) + \frac{A_{xy,\max}}{\kappa_{\mathrm{QP}} K_p^{xy}} + O(\delta), \tag{46}$$

where $A_{xy,\max}$ is the maximum reference acceleration, $\zeta = 1.12$ from Remark 1, and $\kappa_{\mathrm{QP}} = w_t/(w_t + w_e) \approx 0.98$. At canonical $A_{xy,\max} \approx 2.47\,\mathrm{m\,s^{-2}}$, (46) *gives $\bar{e}_{p,xy} \approx 0.27\,\mathrm{m}$, consistent with V1 horizontal RMSE $0.306\,\mathrm{m}$ (gap from centripetal acceleration on curved segments, excluded by the quasi-static derivation).*

Corollary 2 (Robustness to mass, stiffness, drag, and wind)**.** *The envelope* (44) *is robust along four axes: (i) mass mismatch: $T_i^{\mathrm{ff}} = T_i$ uses measured tension, making Lemmas 1 and 2 mass-independent to leading order; (ii) stiffness and drag mismatches: enter through $O(\delta + \eta_{\max})$, absorbed by γ_η without degrading ρ or α_{xy}; (iii) wind: enters through γ_w with steady-state gain $1/K_p^{xy}$. Empirically: P2-B yields $< 1.9\%$ RMSE variation over a 56% mass range with FF active; V2 yields $0.312\,\mathrm{m}$ RMSE under Dryden wind, matching V1 (Corollary 1).*

a) Multi-fault feasibility.: (44) extends to k sequential faults by induction under two conditions: (a) geometric closure $k \leq N-2$, (b) actuator margin $m_L g/(N-k) \leq \kappa_{\mathrm{act}} f_{\max}$ at every step. Canonical $N = 5$, $m_L = 10\,\mathrm{kg}$ admits $k = 2$ with margin; V4 and V5 (dual-fault) yield 3-D RMSE 0.324–$0.328\,\mathrm{m}$, consistent with Theorem 2 on the demonstrated mission family.

b) Empirical validation summary.: Table II maps each claim to its campaign observation; full context is in Section VI.

V. Extension Layers and Theorem Preservation

Three extension layers augment the baseline cascade of Section III and address residual concerns that the identity $T_i^{\mathrm{ff}} = T_i$ does not target: (i) parameter mismatch in (m_L, k_{eff}), (ii) horizon-aware tension-ceiling enforcement, and (iii) post-fault hover-equilibrium asymmetry. Each layer enters the closed loop through a disjoint channel. The L_1 altitude augmentation and the MPC tension-ceiling layer preserve the hypotheses H1–H3 of Theorem 2 without peer communication for control;

TABLE II
EMPIRICAL SUPPORT FOR HYBRID PRACTICAL-ISS CLAIMS.

Claim / Lemma	Campaign	Evidence
Tension cancellation (Lemma 1)	P2-A ablation	$3.6\times$–$4.0\times$ sag increase with FF disabled
Altitude ISS decay (Lemma 2)	V1 nominal	3-D RMSE $= 0.312\,\mathrm{m}$, stable; $\lambda_z = 5.37\,\mathrm{s}^{-1}$ from P_v in (32)
Horizontal ISS decay (Lemma 2)	V1–V5 series	$\alpha_{xy} = 2.38\,\mathrm{s}^{-1}$ from P_{xy} in (32); $\rho = \exp(-\alpha_{xy}\tau_{\mathrm{pend}}) \approx 0.005$
Anti-swing invariance (Remark 1)	V1–V5 series	Post-fault RMSE within 4% of nominal across all variants
Bounded fault jump (Lemma 3)	V3, V4, V5 traces	$\hat{\chi}$ peak 1.1×10^{-4} (V4), 1.3×10^{-5} (V5); qualitatively consistent with the bounded redistribution estimate (37); under equal-sharing hover the canonical coefficient reduces to $\kappa_V\Delta_f^2/(2N_s)$, with quantitative calibration of κ_V deferred to follow-up work
Dwell-cycle contraction $\rho < 1$ (Lemma 4)	V4, V5 dual-fault	$\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$; both satisfy (43) with $c/V(t_k^{\star,-})$ set by the Dryden disturbance floor (Remark 2)
Mass robustness (Corollary 2)	P2-B sweep	1.9% RMSE range over 56% mass variation
Wind robustness (Corollary 2)	V2 wind	V2 3-D RMSE $= 0.312\,\mathrm{m} \approx$ V1

on the demonstrated trajectory family they remain inactive unless their corresponding regime is encountered, with the MPC layer specifically inactive in NF2. The reshape supervisor likewise remains inactive in NF3, but it is governed by the separate result of Theorem 3 and, unlike the other two layers, uses a local detection latch and a $\lceil \log_2 N \rceil$-bit broadcast. This section states the role of each layer and the architectural invariant that keeps the main certificate (44) valid when any combination is enabled. Detailed adaptation laws, MPC formulations, reshape geometry, and full preservation arguments are reported in the supplementary material.

A. L_1-Adaptive Altitude Augmentation

The identity $T_i^{\mathrm{ff}} = T_i$ does not cancel matched disturbances entering through parameter deviations (m_L, k_{eff}). The L_1 layer [13], [14] estimates the residual online and injects an additive altitude correction $u_{\mathrm{ad}}(t)$ between the outer PD and the QP at the injection point of Figure 2. The standard L_1 architecture (Hurwitz reference model with A_m, P_v from Lemma 2, state predictor, projected gradient on $\hat{\delta}_m \in [\delta_{m,\min}, \delta_{m,\max}]$, low-pass filter at $\omega_c = 25\,\mathrm{rad\,s}^{-1}$) is adopted verbatim; the discrete update runs at $T_s = 2\times10^{-4}\,\mathrm{s}$ with adaptation gain Γ. Contractivity of the scalar Euler-discretized update yields the design rule below; the full derivation is in the supplementary material.

Proposition 1 (Scalar L_1 admissible-window bound)**.** *The Euler-discretized L_1 update at step T_s admits two necessary conditions on the adaptation gain Γ:*

$$\Gamma_{\min} \triangleq \frac{\omega_c}{p_{22}} < \Gamma < \frac{2}{T_s\, p_{22}} \triangleq \Gamma^{\star}, \tag{47}$$

expressing bandwidth coverage and discretization stability respectively. For $T_s = 2\times10^{-4}\,\mathrm{s}$, $p_{22} = 0.0210$, $\omega_c = 25\,\mathrm{rad\,s}^{-1}$, this gives $\Gamma_{\min} \approx 1{,}190 < \Gamma = 2{,}000 < \Gamma^{\star} \approx 4.75\times10^5$, placing the canonical operating point $1.68\times\Gamma_{\min}$ above the lower bound with a $238\times$ discretization safety margin.

B. Receding-Horizon MPC with a Penalized Tension-Ceiling Constraint

The single-step QP (26) cannot foresee a tension spike ahead. When safe working load dominates, we replace it with an N_p-step receding-horizon MPC carrying a soft penalized tension-ceiling constraint ($\boldsymbol{s} \geq \mathbf{0}$, weight $w_s = 10^4$). On each tick the MPC propagates a ZOH discretization of the slot-tracking error and a chord-linearized tension prediction, then solves a single warm-started OSQP [23] program; only the first input is applied. Slack penalization preserves unconditional feasibility (Corollary 3); under D1 and Lipschitz-in-parameter right-hand sides, slack activation is $O(\bar{\varepsilon}_q)$ in the parameter-mismatch radius $\bar{\varepsilon}_q = \|q - q^{\star}\|_{\infty}$ (Proposition 2). The MPC reduces to the baseline QP whenever the predicted tension stays clear of the ceiling. Canonical settings: $N_p = 5$ (10 at evaluation), $\Delta t_{\mathrm{MPC}} = 0.01\,\mathrm{s}$, $T_{\max} = 100\,\mathrm{N}$, $k_{\mathrm{eff}} = 2778\,\mathrm{N\,m}^{-1}$.

Corollary 3 (Unconditional feasibility)**.** *The horizon QP is feasible at every tick: $\mathcal{U} = \mathbf{0}$ with $\boldsymbol{s}$ chosen entry-wise as the positive part of the deficit is always admissible.*

Proposition 2 (Slack-activation regime)**.** *Let $q = (m_L, k_{\mathrm{eff}})$, $\|q - q^{\star}\|_{\infty} \leq \bar{\varepsilon}_q$, and let the constraint right-hand side be L_q-Lipschitz in q. Under D1 and an initial tick where the unconstrained QP satisfies the ceiling, $\|\boldsymbol{s}[k]\|_{\infty} \leq L_q\bar{\varepsilon}_q + O(\bar{\varepsilon}_q^2)$.*

C. Fault-Triggered Formation-Reshape Supervisor

After a severance, the survivors retain their pre-fault angular slots, leaving the hover-equilibrium tension distribution asymmetric. This asymmetry is the binding load only when the trajectory is sufficiently low-bandwidth that the dynamic centripetal component does not dominate (NF3, Remark 5). The reshape supervisor reassigns survivors to an equiangular configuration through a C^2 quintic smoothstep over $T_{\mathrm{trans}} = 5\,\mathrm{s}$. Unlike the baseline, this is an *explicitly supervised* layer: each drone runs a local tension-latch detector ($T_{\mathrm{fault}} = 0.5\,\mathrm{N}$, $\tau_{\mathrm{det}} = 100\,\mathrm{ms}$) and a per-fault broadcast of at most $\lceil \log_2 N \rceil$ bits distributes the fault index. The no-detection,

no-communication guarantees of Theorem 2 therefore do not extend to the reshape layer.

Theorem 3 (Globally tension-optimal reshape, $N = 4$)**.** *For $N = 4$ with nominal slots $\phi_i = \pi i/2$ and severed cable $i^\star$, under the symmetric-load-sharing model, the worst-case-tension-minimizing reassignment over the survivors places the diametrically opposite drone at its nominal angle and rotates the two adjacent drones by $\pi/6$ toward the gap. The resulting equiangular 120° spacing minimizes $\max_i T_i$, with closed-form worst-case reduction 25.8%.*

The proof (reflection symmetry, piecewise-quadratic minimax in $\Delta\phi$) and the $V_4 \to V_6$ chained application across $N = 5 \to 4 \to 3$ are reported in the supplementary material.

D. Baseline Theorem Preservation

Each extension modifies the closed loop through a disjoint channel and preserves H1–H3 of Theorem 2 on $\Omega = \Omega_\tau^{\mathrm{dwell}} \cap \Omega_{\mathrm{QP,active}}$. The L_1 layer adds u_{ad} to the altitude component of $\boldsymbol{a}_{\mathrm{target}}$ before the QP, leaving the active set and the identity unchanged; H1, H3 hold verbatim, H2 is controller-independent, and the γ_q-gain in (44) is reduced by the composite L_1 contraction. The MPC layer replaces (26) with the horizon QP of Section V-B: Corollary 3 guarantees feasibility, Proposition 2 bounds the slack, and the closed loop reduces to the baseline whenever the ceiling is loose. The reshape supervisor injects a C^2-bounded slot perturbation through the override hook; H1, H3 hold provided the smoothstep keeps the perturbation in the admissible-reference set, and H2 reduces to a latching rule firing at most once per dwell period. (44) therefore remains valid under any combination of enabled layers; cross-layer interactions are evaluated empirically in Sections VI-F and VI-G.

a) Out-of-scope conditions.: Three regimes lie outside the analysis domain: rapid multi-fault ($\tau_d < \tau_{\mathrm{pend}}$), long slack excursions (run $> 40\,\mathrm{ms}$ or duty $> 2.5\%$), and QP-fallback, violating H2, H1, and H3 respectively. A second class lies outside the architectural category: drone-side faults (rotor loss, brownout), soft cable failures (fraying, attachment slip), tension-sensor failures, and payload-attitude dynamics. These are out-of-category for the architectural thesis, not gaps in the campaign.

VI. Simulation Campaign and Results

All simulations run in Drake [24] with an RK3 continuous-time integrator at a max step of $1\,\mathrm{ms}$. Drones and payload are rigid multibody systems; ropes are bead-chain Kelvin–Voigt spring-dampers whose lumped stiffness is justified by Theorem 1. Wind is body-frame aerodynamic drag per drone and on the payload, using a Dryden low-altitude turbulence model. Table III lists the simulator and plant parameters.

The campaign structure (Table IV) comprises V1–V6 as the pre-declared capability sequence, each adding one stressor, and P2-A–P2-D as targeted ablations. Faults are instantaneous: the rope tension is set to zero; survivors continue without notification.

TABLE III
Simulator and Plant Parameters

Parameter	Value	Unit
Number of drones N	5	—
Drone mass m_i	1.5	kg
Payload mass m_L	10	kg
Rope rest length ℓ_0	1.25	m
Rope stiffness k	25 000	$\mathrm{N\,m^{-1}}$
Formation ring radius	0.80	m
Reference trajectory	3-D lemniscate	—
Horizontal amplitude	3.0	m
Period (nominal)	12	s
Vertical amplitude	0.35	m
Wind model	Dryden low-altitude	—
Mean speed (along $+x$)	4	$\mathrm{m\,s^{-1}}$
σ_u, σ_v	0.8	$\mathrm{m\,s^{-1}}$
σ_w	0.4	$\mathrm{m\,s^{-1}}$
Turbulence seed	42	—
Simulation duration	30	s
Pre-fault cruise window	[8, 12]	s
Actuator thrust ceiling	150	N
MPC tension ceiling (V6 only)	100	N

a) Admissibility-domain audit (H1 and H3).: Theorem 1 and Theorem 2 require three gates to hold on the demonstrated trajectory: H1a (no slack run $> 40\,\mathrm{ms}$), H1b (aggregate slack duty cycle $\leq 2.5\%$), and H3 (QP active-set transition fraction $\leq 1\%$). Table V reports measurements for V1–V6: every variant passes with margin. Worst observed values are $35.8\,\mathrm{ms}$ (H1a; V2–V5, $4.2\,\mathrm{ms}$ margin), 1.66% (H1b; V6, 0.84 pp margin), and 0.10% (H3; V4–V6, 0.90 pp margin). These measurements certify domain membership for the demonstrated mission family but do not prove H1–H3 for all references, wind, or fault schedules; the audit must be repeated for new missions. V6's lower H1a ($31.4\,\mathrm{ms}$) is consistent with MPC damping high-frequency QP transitions.

b) Statistical scope.: All campaigns use fixed Dryden seed 42 (reproducible 2σ exceedance within $[8, 12]\,\mathrm{s}$); reported RMSE, sag, and ablation values are deterministic point estimates. CRN-paired replication with bootstrap CIs is follow-up work.

c) Scope.: V1–V6 begin slightly slack ($1.17\,\mathrm{m}$ vs. $1.25\,\mathrm{m}$ rest length); pickup engages within the first tick and the evaluation window $[8, 30]\,\mathrm{s}$ excludes the transient.

A. Reduction-Fidelity Audit

Two distinct quantities must be kept apart: the *per-rope shape-mode deviation* $|T_i^{\mathrm{KV}} - T_i^{\mathrm{qs}}|$ that Theorem 1 bounds at $\leq C_1\delta + C_2\eta_{\max} \approx 0.076\,\mathrm{N}$, and the *inter-rope asymmetry* $\varepsilon_i \triangleq T_i^{\mathrm{KV}} - T_{\mathrm{eff}}$ around the surviving-rope mean, which is formation-geometry driven and not bounded by the theorem. Across $[8, 30]\,\mathrm{s}$ on V1–V6, $|\varepsilon_i|$ has RMS 7–13 N (P95 up to $24\,\mathrm{N}$, peaks $\sim 60\,\mathrm{N}$ at the most centripetally-loaded instants); after a fault each survivor carries more load and formation angles differ, so T_i^{qs} varies across ropes by $O(10)$ N. The identity $T_i^{\mathrm{ff}} = T_i$ ensures each drone receives the full asymmetric signal, so the stability analysis need not model the asymmetry analytically (Lemma 1). The domain-gate audit Table V is the primary certificate.

TABLE IV
SIMULATION CAMPAIGN MATRIX. CAPABILITY-DEMONSTRATION VARIANTS V1–V6 AND TARGETED ABLATIONS P2-A THROUGH P2-D.

Tag	Controller	Fault Schedule	Wind	Purpose
Capability Demonstration and Admissibility Audit				
V1	Baseline QP	None	No	Nominal tracking floor; C2 bound check
V2	Baseline QP	None	Yes	Wind-rejection baseline
V3	Baseline QP	Drone 0 at t=12 s	Yes	Single-fault recovery
V4	Baseline QP	Drone 0 at t=12 s; Drone 2 at t=17 s (Δt=5 s)	Yes	Dual fault; dwell margin 2.76 s
V5	Baseline QP	Drone 0 at t=12 s; Drone 2 at t=22 s (Δt=10 s)	Yes	Dual fault; dwell margin 7.76 s
V6	Full stack (MPC+L_1+reshape)	Same as V4	Yes	Full-stack integration; hardest scenario
Targeted Ablations and Operating-Domain Probes				
P2-A	Baseline, FF disabled	V3, V4, V5 schedules	Yes	Feed-forward mechanism evidence (C2)
P2-B	Baseline / +L_1	None	No	Mass-mismatch robustness (C3-L_1)
P2-C	MPC, ceiling sweep	V4 schedule	Yes	Tension-ceiling sensitivity (C3-MPC)
P2-D	Baseline + reshape	V4 schedule, period $T_{\mathrm{ref}} \in \{8, 10, 12\}$ s	Yes	Reshape benefit vs. trajectory period (C3-Reshape)

TABLE V
DOMAIN GATE AUDIT. THRESHOLDS: H1A ≤ 40 ms; H1B $\leq 2.5\,\%$; H3 $\leq 1.0\,\%$.

	H1a (ms)		H1b (%)		H3 (%)	
Variant	Value	Pass	Value	Pass	Value	Pass
V1	21.2	✓	1.34	✓	0.07	✓
V2	35.8	✓	1.52	✓	0.07	✓
V3	35.8	✓	1.52	✓	0.09	✓
V4	35.8	✓	1.52	✓	0.10	✓
V5	35.8	✓	1.52	✓	0.10	✓
V6	31.4	✓	1.66	✓	0.10	✓
Gate	≤ 40		≤ 2.5		≤ 1.0	

TABLE VI
INTEGRATED SYSTEM-LEVEL PERFORMANCE (V1–V6, SEED-42). ACCEPTANCE THRESHOLDS: RMSE ≤ 0.35 m; SAG ≤ 100 mm; TENSION ≤ 120 N.

Variant	RMSE	$\mathrm{Sag}^{\mathrm{peak}}$	$\mathrm{Tension}^{\mathrm{peak}}$	Acceptance
V1	0.312m	—	105.3N	Pass
V2	0.312m	—	104.6N	Pass
V3	0.324m	86.8mm	104.6N	Pass
V4	0.328m	95.2mm	104.6N	Pass
V5	0.324m	90.4mm	104.6N	Pass
V6	0.326m	76.1mm	107.6N	Pass

B. Nominal Tracking Performance

V1 (no wind) yields 3-D RMSE $0.312\,\mathrm{m}$ over $[8, 30]$ s. The bound of Corollary 1 is horizontal-only; V1 horizontal RMSE $0.306\,\mathrm{m}$ matches the predicted $0.27\,\mathrm{m}$ to leading order, with the gap from centripetal acceleration on the curved segments. The 3-D value exceeds the corollary because it accumulates altitude error over the $\pm 0.35\,\mathrm{m}$ vertical excursion, which the horizontal-only derivation excludes.

V2 (Dryden wind, $4\,\mathrm{m\,s^{-1}}$ mean, seed 42) yields RMSE $0.312\,\mathrm{m}$, indistinguishable from V1 at single-seed precision. The PD/QP layer absorbs the turbulence without adaptive augmentation, exercising the wind margin of Corollary 2. Table VI consolidates per-variant RMSE, sag, and tension; Figure 3 shows the V4 payload trajectory in 3-D with the lemniscate reference, fault events, and three formation snapshots.

Table VI reports integrated system-level performance for V1–V6 (deterministic seed-42 estimates). The rightmost column marks each variant against the pre-registered thresholds: 3-D RMSE $\leq 0.35\,\mathrm{m}$, sag $\leq 100\,\mathrm{mm}$, tension $\leq 120\,\mathrm{N}$.

C. Fault Progression, Recovery, and Lyapunov Evidence

We audit Theorem 2 against three scenarios of increasing severity: V3 (single fault), V4 (dual, $\Delta t = 5\,\mathrm{s}$), V5 (dual, $\Delta t = 10\,\mathrm{s}$).

a) V3 — single fault.: Drone 0 severs at $t = 12\,\mathrm{s}$; four survivors respond without fault signals or peer communication. Peak 3-D error $0.337\,\mathrm{m}$, IAE $0.452\,\mathrm{m\,s}$, max sag $85.1\,\mathrm{mm}$. Settle time is below the $1\,\mathrm{ms}$ tick resolution (logged as 0 s): the bounded jump of Lemma 3 at the near-equal-sharing hover point falls below one tick.

b) V4 — dual fault, $\Delta t = 5\,\mathrm{s}$.: Drone 2 severs at $t = 17\,\mathrm{s}$, $5\,\mathrm{s}$ after the first. Peak error $0.446\,\mathrm{m}$, IAE $0.666\,\mathrm{m\,s}$, max sag $79.1\,\mathrm{mm}$, settle time $0.158\,\mathrm{s}$. The second-fault transient decays within the same envelope, consistent with $\rho < 1$.

c) V5 — dual fault, $\Delta t = 10\,\mathrm{s}$.: Extending the gap to $10\,\mathrm{s}$ lets the system fully re-settle. The second fault's IAE is $0.741\,\mathrm{m\,s}$ ($+11\%$ over V4) reflecting higher cruise speed at $t = 22\,\mathrm{s}$, not degraded recovery. Max sag $87.4\,\mathrm{mm}$; sub-tick settle.

d) V6 — full stack.: With MPC $+L_1$+reshape active, first-fault sag is $69.9\,\mathrm{mm}$ (-12% vs V4), consistent with the stack preserving the envelope while delivering incremental benefit.

Figure 4 shows the fault-centered timeline for V4 with FF-on (solid) vs FF-off (dashed) on the same wind seed: (a) tracking-error norm, (b) altitude error exposing the sag differential, (c) Lyapunov proxy $V(t) = \xi^\top P_v \xi$, allowing direct visual inspection of the bounded jump (Lemma 3) and the inter-event contraction (Lemma 4) together with mechanism evidence from the FF ablation. V3, V5, FF-held-constant, and per-drone thrust traces are in the supplementary package. Empirical fits on V4 and V5 give peak jump $\hat{\chi}_{\max} = 1.1{\times}10^{-4}$

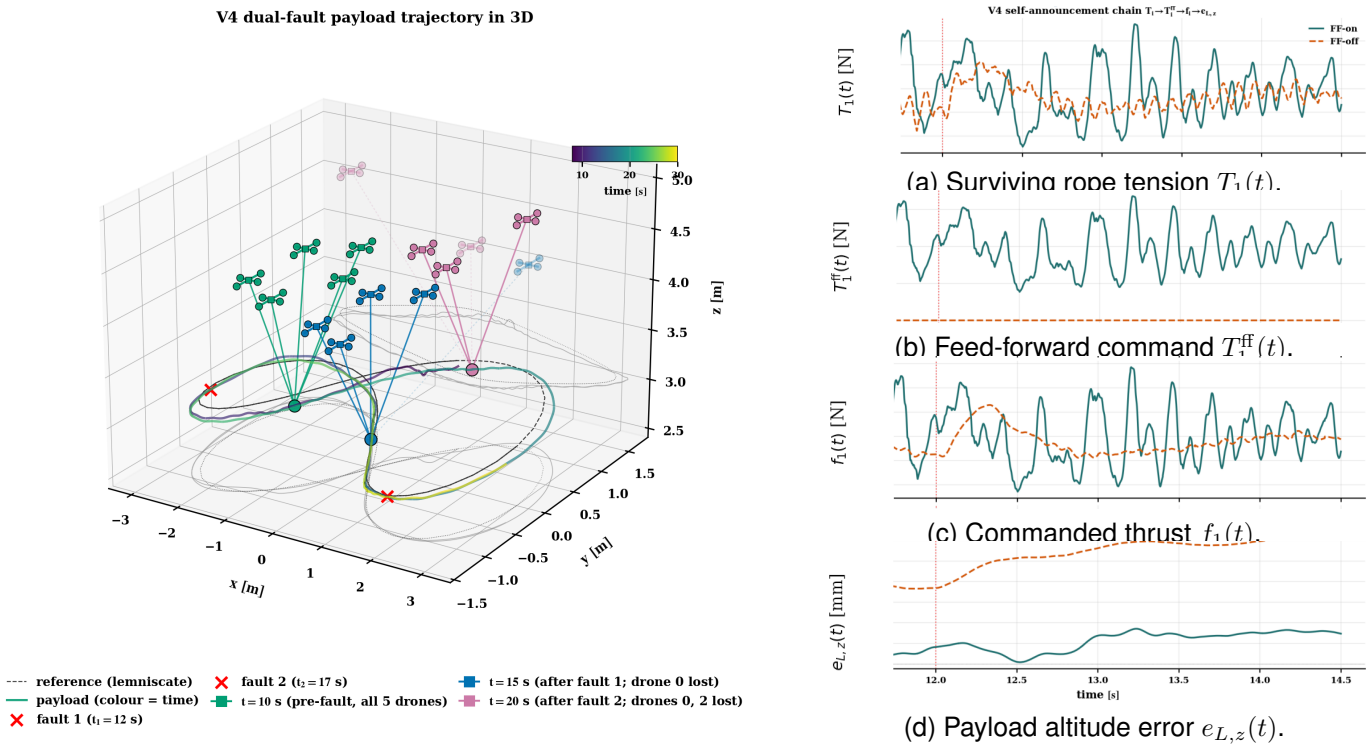


Fig. 3. V4 fault-response visualization. Left: dual-fault payload trajectory over $t \in [8, 30]$ s, with the dashed lemniscate reference, fault events at t_1=12 s and t_2=17 s, and formation snapshots at $t \in \{10, 15, 20\}$ s. Right: self-announcement causal chain for fault 1, where drone 0 is severed at $t = 12$ s. Teal solid denotes the baseline FF-on case, and orange dashed denotes the P2-A FF-off ablation. With the identity active, T_1^{ff} tracks T_1 at the next tick, thrust steps proportionally, and altitude error remains bounded. With the identity disabled, $T_1^{\mathrm{ff}} = 0$ remains flat, the thrust step is absent, and altitude error grows by approximately 300 mm.

(V4), $1.3{\times}10^{-5}$ (V5)—in units of the dimensionless proxy V, not physical units—and dwell-cycle contraction $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$, both well below the $\rho < 1$ sufficiency bound (Table II).

D. Tension Feedforward Ablation

The P2-A campaign provides the strongest quantitative evidence for (28) as the causal recovery mechanism. We disable $T_i^{\mathrm{ff}} = T_i$ in the QP layer and repeat V3, V4, V5.

Without the identity, the altitude channel carries an unrejected load increment, inflating both steady-state floor and post-fault Lyapunov jumps (Lemma 3); Table VII and Figure 4 summarize the 34–39% RMSE increase and 3.6–4.0× peak-sag inflation.

TABLE VII
TENSION FEED-FORWARD ABLATION (P2-A). CRUISE-WINDOW RMSE (M) AND PEAK POST-FAULT ALTITUDE SAG (MM) WITH FF ENABLED AND DISABLED, PER FAULT SCHEDULE.

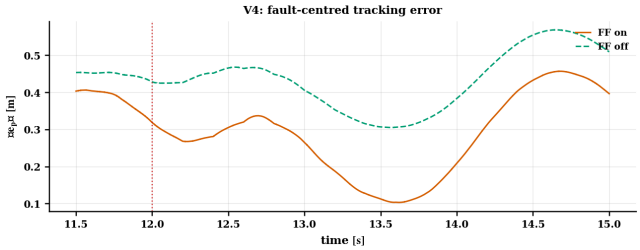
	RMSE (m)			Peak sag (mm)		
Variant	FF on	FF off	Δ%	FF on	FF off	×
V3	0.324	0.432	+34 %	86.8	309	3.6×
V4	0.328	0.457	+39 %	95.2	358	3.8×
V5	0.324	0.444	+37 %	90.4	359	4.0×

E. Dwell-Time Boundary Characterisation

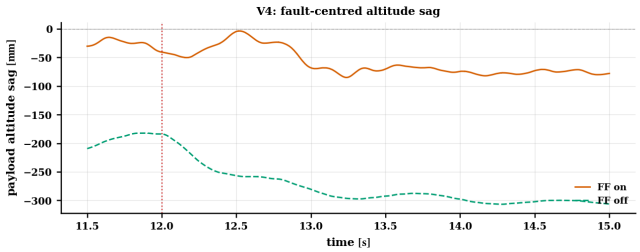
(9) asserts $\rho = e^{-\alpha_{\min}\tau_{\mathrm{pend}}} < 1$ for any $\tau_d > \tau_{\mathrm{pend}}$. V4 and V5 confirm two operating points inside the guaranteed regime but do not characterize behavior as τ_d approaches or falls below τ_{pend}. We close this gap with a boundary sweep on V4 across $\tau_d/\tau_{\mathrm{pend}} \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$ at seed 42.

For each point we compute the per-fault peak altitude excursion $\hat{e}_{p,z}^{\max}(t_k)$ relative to a 0.2-second DC-offset window ending at t_k (removing the post-fault equilibrium shift), then form $\hat{\rho}(\tau_d) = \hat{e}_{p,z}^{\max}(t_2)/\hat{e}_{p,z}^{\max}(t_1)$. Figure 5 reports the result.

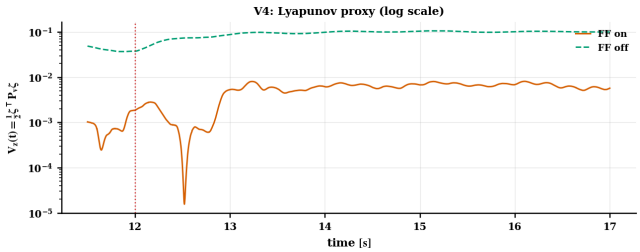
The empirical result is stronger than the hypothesis demands: $\hat{\rho} < 1$ at every tested dwell, including the sub-threshold $\tau_d = 0.5\tau_{\mathrm{pend}}$ where $\hat{\rho} \approx 0.63$. The ratio falls to $\hat{\rho} \approx 0.14$ at $\tau_d = 1.5\tau_{\mathrm{pend}}$. The non-monotonicity at $\tau_d = 2.0\tau_{\mathrm{pend}}$ reflects the trajectory phase the second fault hits (centripetal peak vs trough); a fault-phase sweep would resolve it. The fault-2 peak is consistently below fault-1 across the sweep: the identity bounds the Lyapunov jump at each fault event (Lemma 3), and the inter-event contraction at rate α_{xy} (Lemma 2)—inducing the norm-level recovery rate $\lambda = \alpha_{xy}/2 = 1.19\,\mathrm{s}^{-1}$ of Theorem 2—is sufficient to dissipate that bounded jump within each dwell interval, including the sub-threshold cases. H2 remains a structurally motivated bound; the empirical map shows contraction margin persists below it on this trajectory family.



(a) Tracking-error norm (V4).



(b) Payload altitude error (V4).



(c) Lyapunov proxy $V = \xi^\top P_v \xi$ (V4).

Fig. 4. Fault-centered causal ablation on V4, FF-on (solid) vs FF-off (dashed, P2-A) at seed 42. *(a)* tracking-error norm; *(b)* altitude error; *(c)* Lyapunov proxy $V(t) = \xi^\top P_v \xi$, exhibiting the bounded jump of Lemma 3 and the inter-event contraction of Lemma 4.

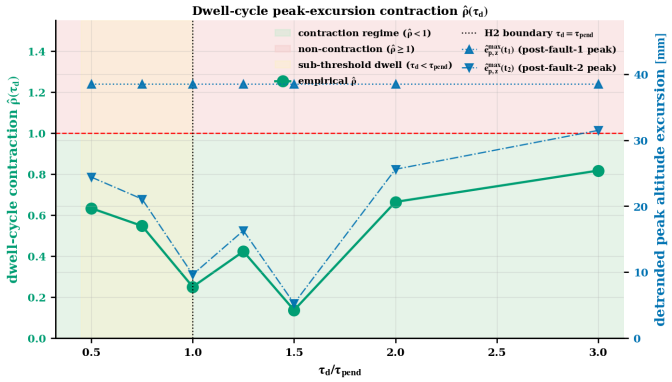


Fig. 5. Dwell-cycle peak-excursion contraction $\hat{\rho}(\tau_d)$ versus normalised dwell $\tau_d/\tau_{\text{pend}}$, V4 schedule, seed 42. Horizontal dashed: $\hat{\rho} = 1$; vertical dotted: H2 boundary. $\hat{\rho}$ remains below unity across the full sweep, including sub-threshold dwell.

F. Parameter Mismatch and Adaptive Augmentation

P2-B evaluates robustness to payload-mass mismatch and the L_1 benefit (Section V-A, C3-L_1). We sweep $m_L \in \{2.5, 3.0, 3.5, 3.9\}$ kg (the PD design range, distinct from the 10 kg V1–V6 payload) with $q^\star$ fixed at 3.0 kg and compare FF-on, FF-off, FF-off+L_1. All runs use the fault-free trajectory to isolate mismatch effects. The L_1 evidence is obtained in rescue mode (FF off, L_1 alone), which isolates the L_1 contribution; main-mode evaluation at the canonical 10 kg point is follow-up work.

FF-on is robust across the range: RMSE varies $< 1.9\%$, confirming the identity pre-compensates the per-drone load change without explicit mass estimation (Corollary 2(i)). FF-off degrades approximately linearly, reaching 0.350 m RMSE at $m_L = 3.9$ kg vs 0.341 m at the nominal. FF-off+L_1 recovers $\sim 40\%$ of the mismatch-induced sag.

The single-axis sweep of Figure 6 is one projection of the four-axis claim of Corollary 2. The full response surface over $(m_L, k_s, c_{\text{drag}}, \bar{w})$ on $n = 100$ CRN-paired seeds is identified as a campaign extension.

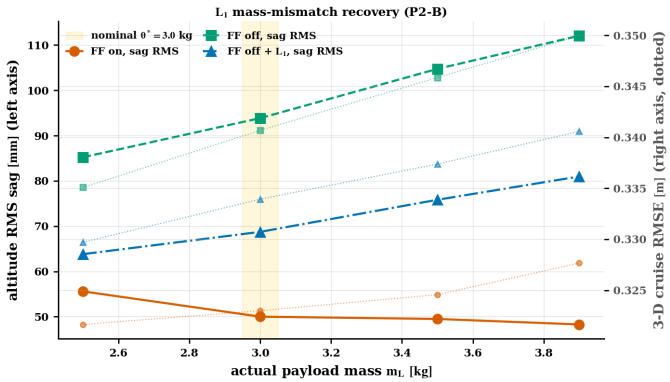


Fig. 6. P2-B payload-mass-mismatch sweep. Cruise-window 3-D RMSE versus $m_L \in \{2.5, 3.0, 3.5, 3.9\}$ kg with $q^\star = 3.0$ kg for FF-on, FF-off, and FF-off+L_1. FF-off+L_1 recovers $\sim 40\,\%$ of the FF-off-minus-FF-on sag gap.

a) L_1 adaptive-gain stability map.: A sweep of $\Gamma \in \{0.5, 2, 10, 30, 50, 80\} \times 10^3$ in rescue mode ($m_L = 3.9$ kg, $q^\star = 3.0$ kg, FF off, no faults) confirms the admissible window of Proposition 1. Altitude RMSE peaks at $\Gamma = 500 < \Gamma_{\min}$ (101 mm), reaches a minimum of 88.5 mm at $\Gamma \approx 10^4$ inside the window, and rises mildly to 91.4 mm at $\Gamma = 8 \times 10^4$. The unimodal shape with worst performance below $\Gamma_{\min}$ empirically supports both window bounds; no instability is observed across the swept range.

G. Extension Characterization and Binding-Regime Evidence

P2-C and P2-D probe the MPC tension-ceiling (C3-MPC) and reshape supervisor (C3-Reshape) on V4. Both yield null findings on the canonical family, complemented by binding-regime probes for the operating conditions under which each extension activates.

a) P2-C — MPC tension ceiling.: We sweep $T_{\max} \in \{60, 70, 80, 90, 100\}$ N against the unconstrained baseline (Table VIII).

Remark 4 (NF2: MPC ceiling non-binding in canonical regime)**.** *Peak tension during the fault transient is dominated by the pre-fault dynamic spike from the lemniscate's centripetal acceleration ($\approx 42\,\mathrm{N}$ on survivors immediately before the first fault, resolving within one pendulum period); this spike is independent of the ceiling. The ceiling targets the post-fault quasi-static regime: at the 5→3 transition with $10\,\mathrm{kg}$ payload, $m_L g/3 \approx 32.7\,\mathrm{N}$, well below all tested ceilings. Lowering the ceiling from $100\,\mathrm{N}$ to $60\,\mathrm{N}$ produces no measurable change (NF2). The ceiling becomes binding when the post-fault load approaches $T_{\max}$—heavier payload, lower ceiling, or fewer survivors. Proposition 2 and Section V-D preserve Theorem 2 whether or not the ceiling is binding.*

b) P2-D — Formation reshape.: We sweep $T_{\mathrm{ref}} \in \{8, 10, 12\}$ s with and without reshape on V4, plus the aggressive-period binding-regime probe at $T_{\mathrm{ref}} = 6\,\mathrm{s}$ (E8). Peak post-fault rope tension is identical with reshape on and off at every tested period: $330.2\,\mathrm{N}$, $191.4\,\mathrm{N}$, $153.8\,\mathrm{N}$, $87.9\,\mathrm{N}$ at $T_{\mathrm{ref}} = 6, 8, 10, 12$ s respectively.

Remark 5 (NF3: Reshape benefit zero across periods and aggressive-period probe)**.** *The reshape supervisor targets the quasi-static-hover tension distribution under the symmetric-load-sharing model of Theorem 3. At the tested periods, centripetal acceleration $a_c = (2\pi/T_{\mathrm{ref}})^2 A \approx 0.82$–$3.29\ \mathrm{m\,s^{-2}}$ continuously modulates the per-rope distribution, and this dynamic modulation dominates the hover-equilibrium asymmetry the supervisor targets—so high centripetal load, far from making the asymmetry binding, makes it* non-*binding. The quasi-static assumption of Theorem 3 therefore fails to bind at any tested period, including the aggressive $T_{\mathrm{ref}} = 6\,\mathrm{s}$ probe (E8) for which the linearised theorem a priori predicted a $\sim 25.8\%$ reduction; the measured benefit there is 0.0%. Pushing reshape into a binding regime would require a lower-bandwidth reference family or an extended objective accounting for the dynamic load. Theorem 3 and Section V-D preserve the envelope outside the binding regime.*

TABLE VIII
MPC TENSION-CEILING SWEEP (P2-C, NF2). PEAK ROPE TENSION AND FRACTION OF MISSION TIME OVER THE COMMANDED CEILING $T_{\max}$, V4 FAULT SCHEDULE.

Ceiling $T_{\max}$	Peak T	Peak post-fault	Time over ceiling
60N	104.6N	87.9N	17.52%
70N	104.6N	87.9N	6.50%
80N	104.6N	87.9N	1.60%
90N	104.6N	87.9N	0.34%
100N	104.6N	87.9N	0.12%
Baseline	104.6N	87.9N	—

A trajectory-phase sweep on V4 ($t_1 \in \{8, 10, 12, 14, 16\}$ s) shows peak post-fault sag varying $12\,\mathrm{mm}$–$55\,\mathrm{mm}$ and 3-D cruise RMSE within $\sim 7\%$; the canonical $t_1 = 12\,\mathrm{s}$ sits at the worst-case end, strengthening rather than weakening the recovery claim. A sub-threshold dwell probe at $\tau_d = 0.5\,\tau_{\mathrm{pend}}$ on the same V4 schedule remains bounded with $\hat{\rho} < 1$ (Figure 5), showing contraction margin beyond H2 sufficiency on this family.

H. Actuator-Margin and Recovery-Time Audit

A reviewer concern with cascade architectures is that the controller may rely on actuator saturation to absorb post-fault loads. We audit this with the per-drone maximum commanded thrust ratio $f_i(t)/f_{\max}$ over $[8, 30]\,\mathrm{s}$ for V3–V6. The variant maxima are 0.73 on V3, V4, V5 and 0.79 on V6, all attained on drone 4, with zero cruise-window time above $0.9\,f_{\max}$ on every variant. The V3–V5 maxima coincide because the three variants share the first-fault schedule (drone 0 severs at $t = 12\,\mathrm{s}$) and the simulations are deterministic on a common wind seed: the variant peak is drone 4's transient at $t = 12.68\,\mathrm{s}$, $0.68\,\mathrm{s}$ into the fault-1 recovery, and the second-fault spike (V4: $t = 17\,\mathrm{s}$, V5: $t = 22\,\mathrm{s}$) lands on a different drone with the trajectory at a different phase and stays below the fault-1 peak on drone 4. V6 reaches 0.79 because the full-stack extensions (L_1 + MPC + reshape) shift the altitude command. Recovery is not actuator-limited on any variant.

a) Spike-then-relax dynamics.: On V4, the three survivors show a 1.18–$1.79\times$ peak-over-settled overshoot decaying within $\sim \tau_{\mathrm{pend}}$, with steady-state thrust per survivor rising $+25\%$ to $+34\%$ over the pre-fault baseline. The survivor sum at $t \in [27, 30]\,\mathrm{s}$ settles near $179\,\mathrm{N}$, balancing $m_L g + 3m_{\mathrm{drone}} g \approx 142\,\mathrm{N}$ plus centripetal/tilt overhead—an independent actuator-channel check of post-fault load redistribution.

Table IX reports per-fault recovery metrics beyond cruise RMSE: peak error norm, peak sag, recovery time t_{rec} to within $\varepsilon = 0.35\,\mathrm{m}$ for $0.3\,\mathrm{s}$ continuous, and IAE over τ_{pend} post-fault.

TABLE IX
PER-FAULT RECOVERY METRICS. t_{rec} THRESHOLD $0.35\,\mathrm{m}$ (3-D, $0.3\,\mathrm{s}$ CONTINUOUS HOLD); IAE INTEGRATED OVER τ_{pend} OR THE GAP TO THE NEXT FAULT, WHICHEVER IS SHORTER.

Variant	Fault	$t^\star$	peak $\lVert e \rVert$	peak sag	t_{rec}	IAE
V3	1	12.0s	481mm	85mm	0.00s	0.52m·s
V4	1	12.0s	481mm	85mm	0.00s	0.52m·s
V4	2	17.0s	457mm	89mm	0.32s	0.70m·s
V5	1	12.0s	481mm	85mm	0.00s	0.52m·s
V5	2	22.0s	469mm	90mm	1.58s	0.81m·s
V6	1	12.0s	481mm	73mm	0.00s	0.51m·s
V6	2	17.0s	460mm	72mm	0.33s	0.70m·s

VII. DISCUSSION

The campaign of Section VI establishes a single architectural commitment: the announcement-detection-reconfiguration timeline is unnecessary when the structural transition self-announces through actuator-side measurements, and the identity $T_i^{\mathrm{ff}} = T_i$ suffices. The five-drone, ten-kilogram dual-fault scenario is one point on the feasibility surface defined by (19).

A. Audit of the Two Main Contributions

Theorem 1 (C1) is defended in two layers. The domain-gate audit (Table V) certifies all six missions sit inside $\Omega_\tau^{\mathrm{dwell}}$ with non-trivial margin. The fidelity layer (Section VI-A) separates the shape-mode deviation $|T_i^{\mathrm{KV}} - T_i^{\mathrm{qs}}|$ (bounded by

Theorem 1 at $\approx 0.076\,\mathrm{N}$ per rope) from the inter-rope tension asymmetry $|\varepsilon_i|$ (RMS $12.4\,\mathrm{N}$ on dual-fault variants, formation-geometry driven, not bounded by the theorem). The identity $T_i^{\mathrm{ff}} = T_i$ absorbs the asymmetry per-drone without centralized averaging.

Theorem 2 (C2) is supported by three converging measurements: all six variants pass the pre-registered thresholds with dual-fault peak sag $95.2\,\mathrm{mm}$ vs $100\,\mathrm{mm}$ ceiling (Table VI); fits on V4/V5 Lyapunov proxy give $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$; the dwell sweep shows $\hat{\rho} < 1$ across $\tau_d/\tau_{\mathrm{pend}} \in [0.5, 3.0]$ including the sub-threshold point.

The simulation audit further characterizes extension layers (L_1, MPC, reshape) through targeted sweeps. The L_1 adaptive-gain sweep shows altitude RMSE is highest at $\Gamma = 500 < \Gamma_{\min}$ (bandwidth-coverage violation) and drops smoothly into the window, reaching a minimum at $\Gamma \approx 10^4$ before mild recovery at $\Gamma = 8 \times 10^4$—the unimodal shape with worst performance at the below-$\Gamma_{\min}$ point is the empirical signature of the two-sided admissible window (47). Both main contributions converge on a single causal anchor: the feed-forward ablation (P2-A) raises cruise RMSE by 34–39% and inflates sag 3.6–4.0× (Table VII and Figure 3), consistent across V3, V4, V5; full CRN-paired replication is deferred.

B. What the Architecture Changes

Existing decentralized cooperative-transport architectures address fault tolerance through one of three families. Active fault-tolerant pipelines [19] estimate, classify, and reassign on a detection-confirmation-reassignment timeline comparable to or longer than τ_{pend}. Passive force-control and leader–follower architectures [20], [25] avoid inter-UAV exchange but presuppose fixed cable cardinality, with no guarantee across the constraint-set transition. Centralized and synchronous-distributed force allocation [1], [4]–[7] is structurally inapplicable: the per-tick global-state synchronization it requires is precisely what the constraint-cardinality transition breaks at the actuator-tick rate.

The commitment converts the transition directly to thrust via $T_i^{\mathrm{ff}} = T_i$, eliminating the detection-reconfiguration timeline rather than substituting a faster version. The empirical signature: $0.00\,\mathrm{s}$ recovery on first faults (sub-tick) and $0.32\,\mathrm{s}$, $0.33\,\mathrm{s}$ on V4/V6 second faults, well inside one pendulum period (Table IX). The four architectural properties of the introduction follow as consequences: *locality* by $\mathcal{I}_i$ (11); *passivity* by the absence of detector, classifier, or scheduler in the loop; *robustness to multi-agent failure* by induction on the fault counter, with the demonstrated $(N{=}5, m_L{=}10\,\mathrm{kg}, F{=}2)$ consuming 27% of the actuator envelope (19); and *team-size independence*, since the per-drone law and lemma stack carry no N structurally and the dependence concentrates in $m_L g/N$ and $m_L g/(N{-}F)$.

The two null findings (NF2, NF3) are architectural evidence, not failed claims. MPC shows identical peak tension $\approx 104.6\,\mathrm{N}$ across all ceilings because the pre-fault dynamic spike dominates the targeted quasi-static load (Table VIII). The reshape supervisor shows 0.0% benefit because centripetal load dominates the hover asymmetry Theorem 3 targets (Remark 5). Both confirm the conventional active-FT machinery is *structurally inactive* on the demonstrated family; Section V-D preserves Theorem 2.

C. Boundary of the Architectural Category

Three regimes lie outside the analysis domain: rapid multi-fault ($\tau_d < \tau_{\mathrm{pend}}$) voids Lemma 4; long slack excursions exit $\Omega_\tau^{\mathrm{dwell}}$; QP fallback breaks D1. A multi-fault extension of the dwell argument—closing all three analytically—is the natural completion of the boundary characterisation.

A second class targets linearisation residuals: tube-MPC folding the recovery-envelope overshoot into the QP RHS would close P2-C; extended-objective reshape would close P2-D; main-mode L_1 at $10\,\mathrm{kg}$ would complete C3-L_1. A third class lies outside the architectural category—drone-side faults, soft cable failures, tension-sensor failures, payload-attitude dynamics—enumerated in Section V-D.

Within this category, the campaign establishes that the identity $T_i^{\mathrm{ff}} = T_i$ delivers hybrid practical input-to-state stability with measured contraction margin across single and dual unannounced severances, that the identity is necessary in the precise sense P2-A isolates, and that the conventional active-FT machinery is structurally inactive on the trajectory family for which the identity is binding.

VIII. Conclusion

The results show that a simple decentralized feed-forward identity enables passive fault tolerance to abrupt cable severance in cooperative aerial transport. By setting each drone's altitude thrust feed-forward equal to its measured cable tension, the controller converts load redistribution into immediate control action using only local measurements. This mechanism removes the need for fault detection, communication, or reconfiguration and operates on the timescale of the plant dynamics.

We analyzed the resulting closed loop as a hybrid system and established a practical input-to-state stability result under structural fault transitions. The analysis produced an explicit recovery envelope with exponential decay and per-fault-cycle contraction and guaranteed bounded tracking error under actuator and dwell-time conditions. A reduced-order model supported the analysis and bounded the effect of cable dynamics on the closed loop.

High-fidelity multibody simulations demonstrated that the controller absorbs single and multiple cable severance events within one payload-pendulum period. Feed-forward ablations support the interpretation that the identity is the active recovery mechanism within the tested baseline architecture and showed that extension layers remained inactive on the demonstrated trajectory family.

These results suggest that, on the stated admissibility domain, passive local control laws can certify recovery for self-announcing cable-severance transitions. Future work will extend the analysis beyond the slack-excursion-bounded domain, investigate hardware validation, and explore applications to other multi-agent and contact-rich systems.

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